The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock on the $W(3,3)$ Substrate

Wil Dahn

June 2026 — machine-verified edition

Abstract

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3,3)$: forty points, forty lines, automorphism group with faithful projective subgroup $\mathrm{PSp}(4,\mathbb{F}_3)$ of order 25920 and full Weyl extension $W(E_6) \cong \mathrm{PSp}(4,3){:}2$ of order 51840. The symplectic double cover $\mathrm{Sp}(4,3)$ also has order 51840, but its central involution is invisible on projective objects; the two extensions must not be conflated. The photon carries $W(3,3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine's states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the dual-torsor compiler (three regular copies of both the flat $\mathbb{F}_3^3$ program shell and the noncommutative Heisenberg state shell, with canonical $27+27+27$ triality selected by the Heisenberg center), the oriented control plane (top homology $H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30$, whose conjugate 5-sectors fuse to one Weyl-visible 10-channel), the mirror middleware (a 2160-slot D_{12} transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget 36/40 and contextual fraction 1/10), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon's own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). We also give the recursive engineering law: a level-n holonet has 40^n photonic leaves, $(40^n-1)/39$ $W(3,3)$ instances, and reversible routing diameter bounded by $8n = 8\log_{40} N$, while durable commits use a separate Császár/tomotope clock $T(n) = 4(7^n - 1)$. The runtime contains an exact parabolic packet decoder: the Bell-line stabilizer splits the 1296 compass incidences into $648 + 324 + 162 + 162$, giving the complete binary prefix code $\{0, 10, 110, 111\}$ for mirror, schedule, and two chiral caches, while the dual point stabilizer supplies two persistent 648-slot address sheets. The two cache branches are now resolved internally: each is the same homogeneous space H/C_4, each is a two-sheet cover of one canonical 81-route base H/D_8, and their 324-slot union has full equivariant commutant D_8 acting in 81 four-state fibers. This split address bus is provably distinct from the older non-split ternary $81 \to 162 \to 81$ transport memory. The newest storage layer turns the tomotope cover pathology into an engineering feature: durable storage is cover-indexed by a $\mathbb{Z}_k^3$ fiber over the stable 48-block packet ABI, sentinel-aware compilation can trade commit-time slack for lower $g = 15$ fault energy, the first exact guard band is the arithmetically forced desynchronization at $n = 5$ with remainder $24 = f$, and the Grünbaum–Coxeter Wythoff operations of the tomotope are exactly the packet-growth ladder $4, 12, 24, 48, 96$. No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

Contents

1 Overture: the machine is the geometry is the carrier

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

Principle 1.1 (Identity of the three layers)**.** *Transporting the photon through the mesh* is *applying a gate* is *routing a packet. One physical process; three descriptions.*

The identification rests on seven exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§2.1): the same $W(3,3)$ is drawn by the photon's *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).

2. **Routing–gate identity** (§4): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4, \mathbb{F}_3)$.

3. **Parabolic address–route duality** (§2.3): the point parabolic supplies two 648-slot address sheets, whereas the Bell-line parabolic decodes the compass fabric by the complete prefix code $0, 10, 110, 111$; its cache quarter is an exact D_8-over-81 four-state address fiber. Address and route are the two nodes of the same C_2 building, while the separate ternary transport extension carries temporal state.

4. **Dual-torsor state–program compilation**: the Steinberg memory restricts to both canonical order-27 shell groups as three regular copies. Bell-shell programs use commuting $\mathbb{F}_3^3$ coordinates; point-shell states use Heisenberg 3^{1+2} coordinates; both compile into the same 81 logical qutrits.

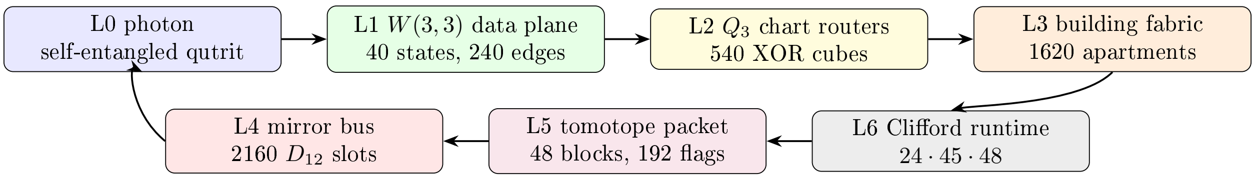
5. **Mirror middleware** (§5): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the C_{12} side but routed by the D_{12} side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.

6. **Universality from optics** (§6.5): the photon's own beam-splitting elements generate the complete Clifford group (symplectic closure $= 51840$, exact), and its matter shell is precisely its magic supply.

7. **Fractal holonet scaling** (§7): recursively replacing every site by a fresh $W(3,3)$ core gives a universal computer/network with exact 40^n leaves and logarithmic reversible routing.

2 The substrate

Definition 2.1 ($W(3,3)$)**.** *Let $V = \mathbb{F}_3^4$ with the alternating form $\langle u, v\rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. The points of $W(3,3)$ are the 40 projective points of V; the lines are the 40 totally isotropic projective lines. Each point lies on 4 lines and each line carries 4 points; collinearity defines the strongly regular graph* $\mathrm{SRG}(40, 12, 2, 4)$.



One loop: carrier → data plane → router → building → runtime → tomotope packet → mirror bus → carrier.

Figure 1: The photonic holonet as an engineering stack. The same photon is data, instruction, and packet; the same finite geometry supplies the network, middleware, and runtime.

Three related symmetry groups occur, and their roles are different:

$$\underbrace{\mathrm{Sp}(4,3)}_{2{\cdot}\mathrm{PSp}(4,3),\ \text{Clifford double cover}}, \qquad \underbrace{\mathrm{PSp}(4,3)}_{25920,\ \text{faithful projective action}}, \qquad \underbrace{W(E_6) \cong \mathrm{PSp}(4,3){:}2}_{51840,\ \text{outer Weyl extension}}.$$

The first and third groups both have order 51840 but are not the same extension: the center of $\mathrm{Sp}(4,3)$ acts trivially on projective points and lines, while the outer involution in $W(E_6)$ acts nontrivially. The projective group acts with permutation rank 3. The ATLAS index-40 subgroups are the two building parabolics $3^{1+2}{:}2A_4$ and $3^3{:}S_4$ [12]. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

Formal grounding (the master formalism). In the master manuscript [1] the whole structure is forced by one Diophantine seed: the *master equation* $q! = 2q$ has the unique positive-integer solution $q = 3$, and the SRG quadratic $x^2 - q!\,x + 2^q = (x-2)(x-4)$ then delivers $\lambda = 2$, $\mu = 4$ and the full parameter closure ($k = 2^q + q + 1$, $v = 40$, multiplicities $f = 24$, $g = 15$, cyclotomic ladder $\Phi_3 = 13$, $\Phi_6 = 7$, $\Phi_{12} = q^4 - q^2 + 1 = 73$, $\Theta = q^2 + 1 = 10$). Two of those closure constants reappear in this paper as *spectral field discriminants*: the chart-web's irrational eigenvalue pairs $1 \pm \sqrt{10}$ and $(-1 \pm \sqrt{73})/2$ (§4) live precisely in $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ — the machine's transport spectrum speaks the same cyclotomic dialect as the physics tier ladder.

2.1 The two carriers and the operator–operand duality

Theorem 2.2 (Two realizations, one geometry)**.** (i) *The* 40 *Witting rays in* $\mathbb{C}^4$ *(the photon's path* $\otimes$ *polarization space) have orthogonality graph isomorphic to the collinearity graph of* $W(3,3)$*.* (ii) *The* 40 *projective classes of two-qutrit Pauli displacements* $D(v) = X^a Z^b \otimes X^c Z^d$ *on* $\mathbb{C}^9$ *(the photon's past* $\otimes$ *future time-bin space) have commutation graph equal to the same* $W(3,3)$*, via* $D(u)D(v) = \omega^{\langle u,v\rangle} D(v)D(u)$*.* Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40 \times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

2.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\mathrm{PSp}(4,3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

index	subgroup	points	lines	vacuum	cubic surface
27	$2^4{:}A_5$	$[40]$	$[40]$	homogeneous (icosahedral)	27 lines
36	S_6	$[40]$	$[10,30]$	spread fibration	36 double-sixes
40	parabolic	$[1,12,27]$	$[4,36]$	holonet point vacuum	trihedra I
40	parabolic	$[4,36]$	$[1,12,27]$	dual line vacuum	trihedra II
45	$(2T{\times}2T){:}2$	$[8,32]$	$[16,24]$	binary polar	45 tritangents

The physical decomposition $40 = 1{+}12{+}27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group O_h ($= S_4 \times \mathbb{Z}_2$, order 48) is the chart symmetry of §4, and the binary icosahedral group $2I = \mathrm{SL}(2,5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits $[20,20]$ and line orbits $[10,30]$ echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808`*–*`bt813`*.*

2.3 The parabolic router: address and route are dual

The two index-40 parabolics are not duplicate descriptions of one local neighborhood. They execute different halves of a packet protocol. This becomes visible only after the compass census: there are two 216-element classes of icosahedral A_5 cores, and exactly $1296 = 216 \cdot 6$ cross-class pairs meet in D_{10}.

Theorem 2.3 (Bell–compass parabolic router)**.** *Fix an isotropic Bell line. Its projective Siegel parabolic* $3^3{:}S_4$ *has four orbits on the* 1296 *incident compass pairs:*

$$162_L + 162_R + 324 + 648.$$

They are distinguished objectwise by the Bell line lying in the pentad core's 5_L*,* 5_R*,* 10*, or* 20 *line orbit. The normalized orbit sizes satisfy the Kraft equality*

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} = 1,$$

so they admit the complete prefix decoder

$$110 \mapsto \textit{left cache}, \quad 111 \mapsto \textit{right cache}, \quad 10 \mapsto \textit{schedule}, \quad 0 \mapsto \textit{dark mirror}.$$

The outer Weyl involution fuses exactly the two 162 *cache orbits, leaving* $324_{\mathrm{cache}} + 324_{\mathrm{schedule}} + 648_{\mathrm{mirror}}$*. By contrast, the dual point parabolic has two regular projective orbits of size* 648*; the outer extension preserves rather than fuses these address sheets.* Witness: `bt859_bell_compass_parabolic_router.py`.

The equality case of Kraft–McMillan means the routing words fill a binary decision tree with no unused branch [13]. Here the code lengths were not optimized from assumed traffic probabilities: they were read from exact group orbits. A packet first asks whether its Bell line is in the dark mirror half; if not, whether it is in the common schedule quarter; only the remaining cache quarter needs the third, chiral bit. The decoder is self-delimiting and constant-depth, with expected word length

$$\frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}$$

under the invariant uniform measure on the 1296 compass pairs.

orbit	Bell-line position	prefix	engineering action
648	dark 20-orbit	0	enter mirror bus
324	common schedule 10-orbit	10	timetable-local route
162	absolute pentad 5_L	110	left cache
162	absolute pentad 5_R	111	right cache

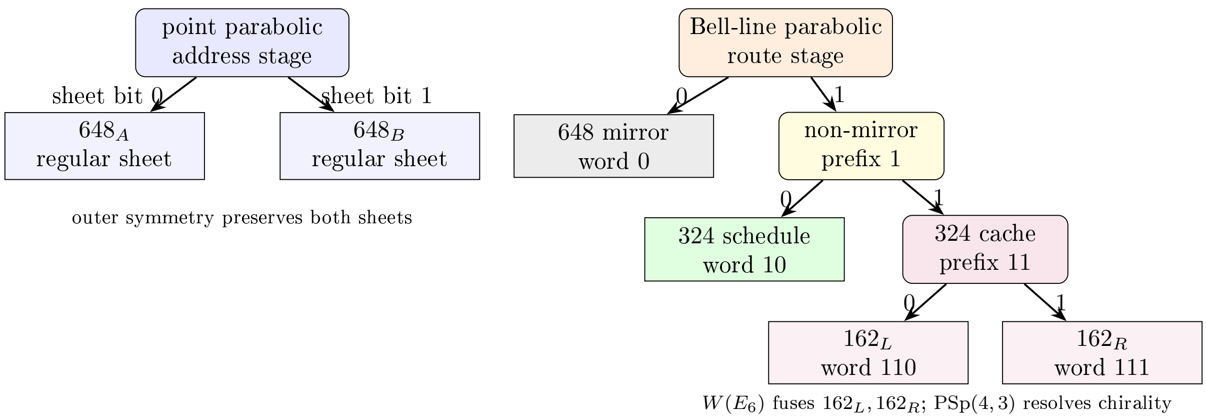


Figure 2: The two C_2 parabolics implement a dual protocol. Point localization selects one of two address torsors; Bell-line localization decodes a complete variable-length route word.

This theorem also closes an honesty boundary. The Bell stabilizer and the compass incidence set both have cardinality 1296, but they are not a regular G-set identification. Projectively, the Bell stabilizer has order 648 and the orbit stabilizers are $1, 2, 4$. The useful result is stronger than the count: a hierarchical decoder and a persistent address bit are forced by the two parabolic actions.

3 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon's own registers.

3.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \mapsto \frac{1}{\sqrt{2}}(|H, a\rangle + |V, b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 2.2(i).

3.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate F_3; a three-bin delay ladder $(0, \tau, 2\tau)$ defines past and future registers; one electro-optic modulator applies $\mathrm{CX}_{p\to f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle \;=\; \mathrm{CX}_{p\to f}\,(F_3\otimes I)|0\rangle_p|0\rangle_f \;=\; \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p|j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any U in the future arm, the recombination visibility is the *trace–Choi witness*

$$V(U) \;=\; \frac{|\mathrm{Tr}\,U|}{3}, \qquad V(F_3)=\tfrac{1}{3}, \quad V(X)=V(Z)=0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness:* `bt820_self_entanglement_protocol.py`.

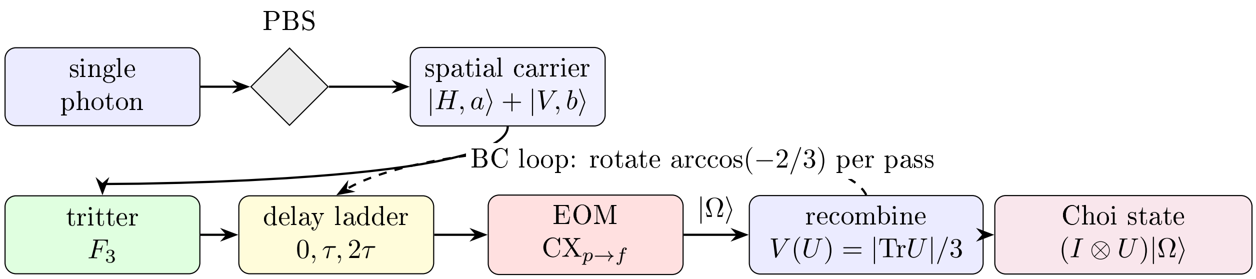


Figure 3: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §11.

4 The network: atlas, building, and routing

Theorem 4.1 (The hypercube atlas)**.** *$W(3,3)$ contains exactly 540 skew line pairs. For each pair the non-collinearity graph on its 8 points is the cube graph Q_3, with full symmetry group O_h of order 48 (these are the only order-48 stabilizers; the binary $2O$ does not occur). Each chart admits an exact $\mathbb{F}_2^3$ Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every $W(3,3)$ nonedge is a chart edge in exactly $12=k$ charts; every edge is an antipode pair in exactly $9=q^2$.* Witnesses: `bt773`, `bt777`, `bt778`.

4.1 Hypercube networking, not hypercube decoration

The cube is not only a picture for one chart. It is the local interconnect primitive. A chart gives eight simultaneously valid addresses

$$000,\ 001,\ 010,\ 011,\ 100,\ 101,\ 110,\ 111 \in \mathbb{F}_2^3,$$

and the elementary network instruction is the dimension-order hypercube hop

$$a \longmapsto a\oplus e_i, \qquad i\in\{1,2,3\}.$$

This is the standard e-cube rule of hypercube networks: compare source and target, flip one differing coordinate at a time, and finish in at most three local hops [6]. In the holonet that same XOR flip is also a register operation, because the chart address is a finite "which-register" word on the photon's self-entangled carrier. The local routing primitive is therefore simultaneously

$$\text{bit flip} \quad = \quad \text{qutrit-frame update} \quad = \quad \text{optical path choice.}$$

A packet in the photonic holonet is not merely a payload. The natural packet header is

$$\mathcal{P} = (\text{payload, Pauli frame, chart word, apartment hop, mirror slot, clock phase}).$$

The payload is the quantum state or teleported gate; the Pauli frame is the two-trit feedforward record; the chart word is the $\mathbb{F}_2^3$ route inside the current cube; the apartment hop selects the next chart; the mirror slot chooses the D_{12} transport bus of §5; and the clock phase is the cyclic C_{12} interrupt that keeps the selector coherent. This is why the same object can be called a computer architecture and a network architecture. A node never sends data to a processor: the act of transport is the processor's gate action.

Theorem 4.2 (The fabric is the building)**.** *The chart-adjacency web (*540 *nodes, two charts adjacent when one's axis is a disjoint transversal pair of the other) is* 6*-regular with exactly* 1620 *edges, and these edges are in canonical bijection with the* 1620 *apartments of the Tits building of* $\mathrm{Sp}(4, \mathbb{F}_3)$*. The web has diameter* 5*; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a* 15*-dimensional sentinel eigenspace.* Witnesses: `bt777`, `bt779`, `bt744`.

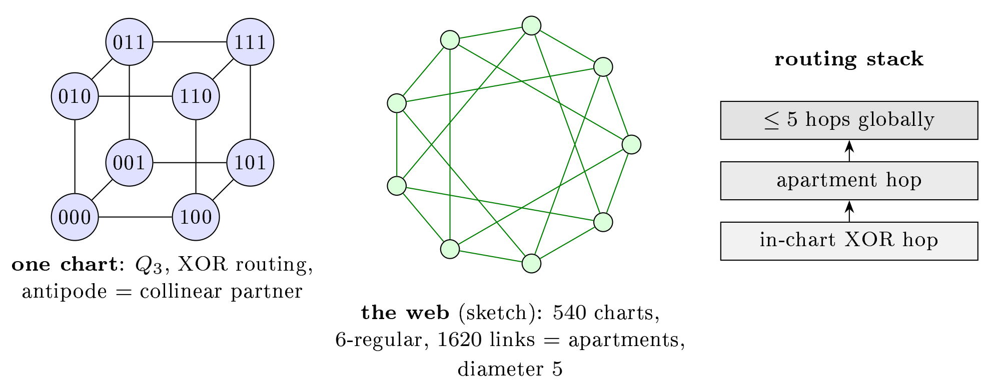


Figure 4: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§6), routing is computation. At scale the recursion is the holonet ladder: 40^n cores at level n, the same $\mathrm{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

Corollary 4.3 (Two-plane routing bound)**.** *Inside a chart the worst route length is* 3*, the diameter of* Q_3*. Between charts the worst route length is* 5*, the diameter of the apartment web. One recursive digit of a holonet address therefore costs at most*

$$3 + 5 = 8$$

reversible transport moves. BT827 globalizes this to $8n = 8 \log_{40} N$ *for* $N = 40^n$ *photonic leaves.*

5 The middleware: tomotope mirrors and the runtime bus

Definition 5.1 (Tomotope packet). *In this paper the tomotope is used in the only way the finite calculation currently justifies: as a local middle-layer incidence packet. A tomotope packet has four vertex carriers, twelve local edge axes, sixteen triangular face labels, and forty-eight edge–face incidence blocks. The doubled base/shadow fiber gives* 192 *flags. Thus the tomotope is the durable packet format sitting between the fast cube router and the global mirror bus; it is not introduced as an ambient continuous torus, and it is not identified with the rectangle clock.*

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is C_{12} and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is D_{12} and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

Theorem 5.2 (The D_{12} mirror bus). *The* 540 *cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has* $540 \cdot 4 = 2160$ *slots. The map*

$$(\textit{chart, common transversal line}) \longmapsto (\textit{chart, antipode pair cut out by that transversal})$$

is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order 12 *and GAP identifies it as* D_{12}*, not* C_{12}*.* Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the tomotope middle layer. Each transversal is a K_4, and each K_4 has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \qquad 4 \cdot 3 = 12 \text{ edge labels}, \qquad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

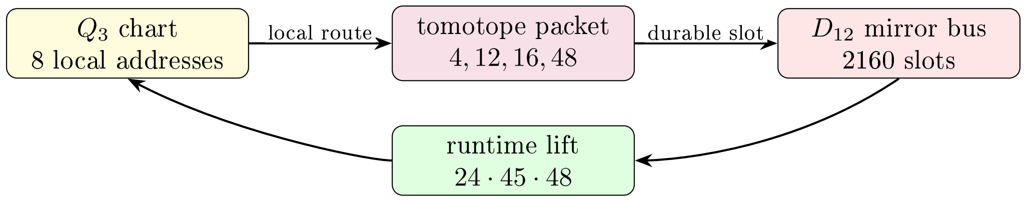
edge–face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3 \rangle$ tomotope block layer, not an analogy to a toroidal polyhedron. *Witness:* *`bt814_tomotope_middle_layer_from_residual_tetrahedra.py`*.

Theorem 5.3 (Photonic mirror middleware). *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

Here $24 = f$ *is the full* $\mathrm{Sp}(4,3)$ *lift of the projective* D_{12} *mirror-slot stabilizer,* 45 *is the hyperbolic polar-pair/tritangent geography, and* 48 *is the local tomotope edge–face middle layer (also the order of the chart group* O_h*). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomotope middle carrier.* Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomotope local block. In software terms, C_{12} is the clock interrupt, D_{12} is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.



fast XOR routing is committed through the tomotope middle layer and reflected by the mirror bus.

Figure 5: Middleware schematic. The cube is the reversible local router; the tomotope is the edge–face packet format; the D_{12} bus is the global mirror transport object.

6 The software: braids, teleported gates, universality

6.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual ($\pm$) encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`, `bt741`.

6.2 The photon is its own gate

A gate U is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs U (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`.

6.3 Instruction set architecture

The practical instruction set is finite:

$$\mathcal{I}_{\text{holo}} = \{F_p, F_f,\ S_p, S_f,\ \mathrm{CX}_{p\to f},\ \sigma^5 = Z,\ D_{12}\text{-mirror},\ M_{36}\text{-magic}\}.$$

F and S generate single-qutrit Clifford frame changes; CX couples past to future; $\sigma^5 = Z$ gives exact braid-register flips; the D_{12} mirror operation moves a packet through the atlas without confusing it with the cyclic selector clock; and M_{36} denotes injection from the thirty-six magic rays of §9. The Clifford part normalizes Pauli frames and is efficiently classically trackable; it is not universal by itself. Universality starts only when the Clifford runtime can consume the intrinsic magic sector. This is the engineering meaning of "matter equals magic": the same thirty-six rays that appear as the matter shell are the non-Clifford fuel for the processor.

6.4 The minimal classical shadow

The smallest classical headline compatible with the substrate is the pair

$$(\lambda, q) = (2, 3).$$

This is the same two-state, three-symbol signature made famous by the Wolfram–Smith weakly universal Turing machine. The present paper does not use that result as the proof of quantum universality; the proof is the Clifford closure plus magic/contextuality [8, 7]. The point is sharper: the classical shadow of the photonic holonet lands on the same minimality boundary. The quantum machine has forty rays and an $81 = q^4$ protected register, but its smallest visible finite-state control alphabet is already the substrate pair $2, 3$.

6.5 The Universality Theorem

Theorem 6.1 (Clifford completeness from optics)**.** *The symplectic images of the photon's physical elements — the tritter F_3 on either register, the quadratic phase plate S, and the delay-conditioned* CX *— generate all of* $\mathrm{Sp}(4, \mathbb{F}_3)$ *(closure of order* 51840*, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases.* Witness: `bt825_universality_theorem.py`.

Theorem 6.2 (Universality)**.** *Clifford completeness, together with the machine's intrinsic magic supply (§9) and the strict positivity of its contextual fraction (*$1/10$*, exact), implies universal quantum computation by magic-state injection; for qutrits, contextuality is necessary and sufficient for magic distillation (Howard–Wallman–Veitch–Emerson) [3]. One self-entangled photon on the $W(3, 3)$ mesh is a universal quantum computer whose transport is its gate action.*

7 Fractal scaling: the computer is the network

The single-core theorem is not the end of the architecture. The substrate is self-similar: any node may be replaced by a fresh copy of the same forty-node holonet, with the parent line/point incidence becoming the routing grammar for the child layer.

Definition 7.1 (Recursive holonet)**.** *Let $\mathcal{H}_0$ be one self-entangled photonic qutrit. Define $\mathcal{H}_n$ recursively by replacing each point of one copy of $W(3, 3)$ with a copy of $\mathcal{H}_{n-1}$. Thus $\mathcal{H}_1$ is the single-core machine of this paper and $\mathcal{H}_n$ is a* 40*-ary fractal computer/network.*

Theorem 7.2 (BT827 holonet scaling law)**.** *A level-n holonet has*

$$N_n = 40^n$$

photonic leaf cores and

$$I_n = 1 + 40 + \cdots + 40^{n-1} = \frac{40^n - 1}{39}$$

total $W(3, 3)$ instances. Its total edge-qutrit slots are $240 I_n$, its directed transport slots are $480 I_n$, its chart routers are $540 I_n$, its apartment links are $1620 I_n$, its mirror slots are $2160 I_n$, and its local Clifford runtime atoms are $51840 I_n$. Worst-case reversible routing across the recursive address uses at most

$$8n = 8 \log_{40} N_n$$

moves, where $8 = 3 + 5$ is one in-chart hypercube diameter plus one chart-web apartment diameter. Durable commits use the distinct Császár/tomotope clock

$$T(0) = 1, \qquad T(g) = 4(7^g - 1) \quad (g \geq 1).$$

Witness: `bt827_holonet_fractal_architecture.py`.

level n	leaves 40^n	$W(3,3)$ instances	mirror slots	runtime atoms	route bound
1	40	1	2160	51840	8
2	1600	41	88560	2125440	16
3	64000	1641	3544560	85069440	24
4	2560000	65641	141784560	3402829440	32

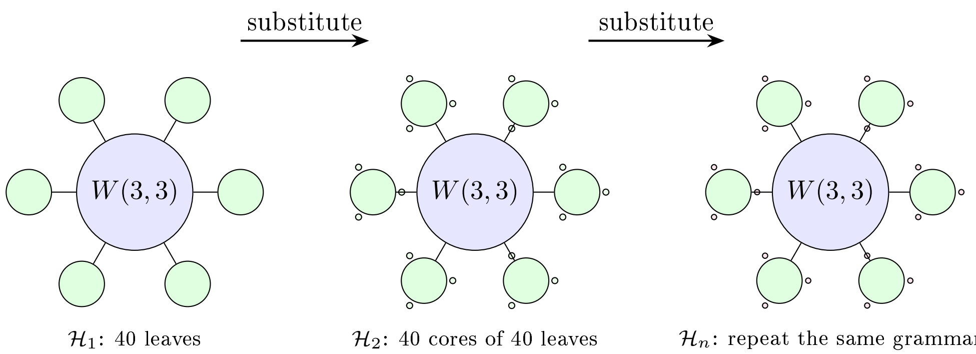


Figure 6: Fractal scaling schematic. The drawn fans show only a few children; the rule is exact substitution of forty child holonets at each substrate point. Routing grows logarithmically in the number of photonic leaves because each recursive digit costs at most eight reversible moves.

This resolves the computer-engineering question. A conventional machine grows by adding processors connected by a separate network. The holonet grows by adding copies of the same object; each copy contains its own processor, router, memory, clock, and packet format. There is no von Neumann bus to saturate. At every scale, the edge set is both communication bandwidth and entangling-gate availability, the chart atlas is both routing table and instruction decoder, and the Steinberg 81-sector is both protected memory and fault-tolerant logical space.

8 Runtime closure: compile, monitor, commit

The architecture now has an executable operating-system loop.

$$\text{address word} \longrightarrow \text{packet route} \longrightarrow \text{sentinel monitor} \longrightarrow \text{durable tomotope commit.}$$

BT828–BT831 make each arrow finite and testable.

Theorem 8.1 (BT828 packet compiler)**.** *A recursive address word with digits in* $\{0,\ldots,39\}$ *compiles to a sequence of digit packets*

$$(Q_3\text{-}XOR\ hops,\ apartment\ hops,\ D_{12}\ mirror\ slot,\ C_{12}\ phase,\ tomotope\ block).$$

Each digit uses at most three XOR hops and at most five apartment hops, hence at most eight reversible moves. The stress route

$$(0,7,14,21,28,35) \longrightarrow (39,32,25,18,11,4)$$

compiles to 47 *reversible moves, below the level-six bound* 48. Witness: `bt828_holonet_packet_compiler.py`.

Theorem 8.2 (BT829 sentinel monitor)**.** *The projector onto the $g = 15$ adjacency sector of $W(3,3)$ has exact entry profile*

$$P_{-4}(x,x) = \frac{3}{8}, \qquad P_{-4}(x,y) = \begin{cases} -\frac{1}{8}, & x \sim y, \\ \frac{1}{24}, & x \nsim y. \end{cases}$$

Consequently a full context line K_4 is sentinel-invisible, while localized, mirror, and shell faults activate the sentinel with exact energies:

fault	*energy*	*normalized fraction*
single point	3/8	5/13
adjacent edge	1/2	5/19
nonedge mirror pair	5/6	25/57
gauge shell $N(x)$	6	5/7
matter shell	27/8	5/13

The immune system is therefore not generic noise detection: it is blind to legal context-line traffic and tuned to off-context routing and shell congestion. Witness: `bt829_fault_sentinel_monitor.py`.

Theorem 8.3 (BT830 two-phase commit clock)**.** *The runtime has two clocks. The reversible prepare phase finishes in at most $8n$ moves. The durable commit phase waits for*

$$T(n) = 4(7^n - 1).$$

For $1 \leq n \leq 24$, $T(n)$ is always divisible by 24 and by 8, so it is aligned with both the local runtime stabilizer and the single-digit route window. But it is not always divisible by the full $8n$ route epoch: the first desynchronization is $n = 5$, with remainder $24 = f$. Thus the commit layer is deliberately slower and cover-like; it is not merely the route clock renamed. Witness: `bt830_two_phase_commit_clock.py`.

8.1 The infinite-cover warning

The tomotope contributes a harder boundary than the earlier packet picture showed. Monson, Pellicer, and Williams prove that the tomotope has infinitely many distinct minimal regular covers; for coprime odd $p, q > 1$ there are minimal covers R_p, R_q of the tomotope such that neither covers the other, and neither covers the special R_2 cover [11]. They also record the toroidal cover arithmetic

$$|\mathrm{Mon}(Q_k)| = 36864\, k^6, \qquad Q_k : \ (4, 24, 32, 8, 4)k^3$$

for vertices, edges, triangles, tetrahedra, and octahedra.

Architecturally this is a big correction. The 48-block tomotope middle layer is a stable local ABI, but the global regular cover is not unique. Durable storage must therefore carry a *cover index* k as an implementation gauge. Fast routes compile to the same invariant packet, while persistent storage may choose one minimal cover family without invalidating the other non-comparable families. *Witness: `bt831_tomotope_minimal_cover_architecture.py`.*

8.2 Cover-indexed storage, rerouting, and guard bands

The infinite-cover warning would be useless as engineering unless it changed the runtime. BT832–BT834 make it operational. The principle is that the local tomotope packet is the ABI and the chosen regular cover is the storage backend.

Theorem 8.4 (BT832 cover-indexed durable storage). *For every supported cover index k, durable storage is the lifted packet space*

$$\{0, \ldots, 47\} \times \mathbb{Z}_k^3, \qquad |slots| = 48k^3,$$

with projection back to the fast-route ABI by forgetting the $\mathbb{Z}_k^3$ coordinate. The kernel to the base tomotope packet has order

$$k^6 = (k^3)^2,$$

matching the Monson–Pellicer–Williams toroidal cover law $|\mathrm{Mon}(Q_k)| = 36864k^6$. Changing k changes durable capacity and kernel size, but not the packet program seen by the compiler. Tested cover gauges $k = 3, 5, 7, 11, 13$ all reduce exactly to the same 48-block ABI, and larger k reduces the load fraction of the same stress program. Witness: `bt832_cover_indexed_durable_storage.py`.

Theorem 8.5 (BT833 sentinel-aware packet rerouting). *The BT829 projector supplies a compiler-visible cost function. Before the durable commit, each digit route may insert up to two W33 waypoint points and minimize*

sentinel energy first, reversible move count second.

The extra reversible moves are charged to the slower BT830 commit phase, not to the BT827 fast $8n$ route bound. In the level-six stress program the direct route uses 47 moves and total sentinel energy 4; the sentinel-aware route uses 58 moves, still far below the 470592-tick commit window, and lowers the total sentinel energy to 2. Each tested digit route strictly lowers its own $g = 15$ energy; adjacent endpoints can be context-completed to zero, and nonedge mirror pairs drop below their direct $5/6$ activation. Witness: `bt833_sentinel_aware_packet_rerouter.py`.

Theorem 8.6 (BT834 desynchronization guard-band arithmetic). *The two-phase clock has an exact number-theoretic boundary. Full route-epoch synchronization is*

$$8n \mid T(n) = 4(7^n - 1),$$

equivalently

$$2n \mid 7^n - 1.$$

For every odd prime power $p^a \mid n$ with $p \neq 7$, this requires $\mathrm{ord}_{p^a}(7) \mid n$; if $7 \mid n$, synchronization is impossible because the base is not coprime. The first failure is $n = 5$: $\mathrm{ord}_5(7) = 4 \nmid 5$, and the remainder is

$$T(5) \bmod 40 = 24 = f.$$

Thus the first cover-index where durable storage and the full route epoch separate is not empirical drift; it is the $f = 24$ guard band of the tomotope-backed runtime. Witness: `bt834_desync_guard_band_arithmetic.py`.

Theorem 8.7 (BT838 tomotope Wythoff runtime ladder). *The Grünbaum–Coxeter/tomotope operation tables give the local tomotope counts*

$$\textit{original, rectified, truncated, maximal expanded, omnitruncated} \quad = \quad 4,\ 12,\ 24,\ 48,\ 96.$$

These are exactly the runtime packet-growth layers:

$$4 = \mu \quad (\textit{vertex carriers}), \qquad 12 = k \quad (\textit{edge axes}), \qquad 24 = f \quad (D_{12}\ \textit{lift}),$$

$$48 = 2f \quad (\textit{BT814 packet ABI}), \qquad 96 = 4f \quad (\textit{half-flag boundary}).$$

The doubled omnitruncated boundary gives the verified full packet flags, $2 \cdot 96 = 192$. After a cover lift this becomes

$$48k^3 = \textit{BT832 lifted packet capacity}, \qquad 192k^3 = |W_k|,$$

the W_k order already recorded in the tomotope cover architecture. Thus the Wythoff operations are not decorative variants; they are the compiler's packet expansion states:

$$\textit{carrier} \to \textit{edge-axis} \to \textit{mirror lift} \to \textit{packet ABI} \to \textit{flag boundary}.$$

Witness: `bt838_tomotope_wythoff_runtime_ladder.py`.

9 The fuel: matter = magic

Theorem 9.1 (The triality). *In the photon's two-qubit reading, exactly the 4 basis rays are stabilizer states and all 36 ω-rays are magic (no stabilizer state has support 3). Consequently three classifications coincide exactly, on rays $[4, 36]$ and on contexts $\{4{:}1,\ 1{:}12,\ 0{:}27\}$: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata.* The matter shell is the machine's non-classical resource sector. Witness: `bt822`.

Theorem 9.2 (Three grades and the cube inside). *Stabilizer fidelity splits the 36 magic rays into grades*

$$\underbrace{8}_{2^q} \textit{ deep } \left(F = \tfrac{2+\sqrt{3}}{6}\right), \qquad \underbrace{24}_{f} \textit{ mid } \left(F = \tfrac{5+2\sqrt{3}}{12}\right), \qquad \underbrace{4}_{\mu} \textit{ shallow } \left(F = \tfrac{3}{4} = \tfrac{q}{\mu}\right),$$

the shallow grade sitting exactly at the Werner decoherence threshold. The 8 deep rays are precisely the point set of a hyperbolic polar pair (the 24-cell vacuum axis), and the 27 fully-magic contexts stratify by grade signature as $1+8+12+6$ — the cell census of the cube. Witnesses: `bt822`, `bt823`.

Theorem 9.3 (Exact contextuality budget). *The maximum number of contexts a classical $\{0, 1\}$ valuation can satisfy exactly-once is exactly $36 = (q!)^2$ (complete branch-and-bound over all 2^{40} markings): classicality saturates at the spread count, the deficit is exactly $\mu = 4$, and the contextual fraction is exactly $1/10 = 1/\Phi_4$. A perfect valuation would be an ovoid, and $W(3, 3)$ has none: the true independence number is $\alpha = 7 = \Phi_6$, attained only by the beacon heptads of §12, against the Lovász bound $\vartheta = 10$ — the $\vartheta - \alpha$ gap $3 = q$ is the combinatorial face of Kochen–Specker.* Witnesses: `bt818`, `bt823`.

10 Memory, protection, and the immune system

Theorem 10.1 (The code register is the Steinberg module). *The companion paper's CSS code $[[240, 81, 4, 3]]_3$ stores its 81 logical qutrits in H_1 of the $W(3, 3)$ 2-complex (40 vertices, 240 edges, 160 in-line triangles). Complete character computation over all 25920 group elements gives*

$$\langle \chi_{H_1}, \chi_{H_1} \rangle = 1.$$

Therefore $H_1 \otimes \mathbb{C}$ is irreducible *of dimension 81, hence equal to the unique 81-dimensional irreducible — **the Steinberg module** — with $\chi_{H_1}(g) = \#\text{fixflags} - \#\text{fixpoints} - \#\text{fixlines} + 1$ (the Solomon–Tits alternating sum) verified for every g. The QECC logical space and the Schur-protected register below* *are one object: any equivariant logical error is a scalar, so only symmetry-breaking noise can touch the code, and the sentinel/clock layers are exactly the symmetry monitors. The homological dictionary then completes: H_0 is trivial (the vacuum pole), H_1 is Steinberg (the register), and H_2 (dim 40, one tetrahedron-boundary 2-cycle per line) is the* sign-twisted *line module — a fixing symmetry acts on a line's 2-cycle by the parity of its induced S_4 action (verified for all 25920 elements;* not *the plain permutation module): the top homology is the* oriented *timetable carrier, feeding the chirality ledger.* Witnesses: `bt861_code_register_is_steinberg.py`, `bt862_h2_is_the_line_module.py`.

Corollary 10.2 (Three generations from Steinberg vanishing)**.** $\chi_{\mathrm{St}}(g) = 0$ *iff* $3 \mid \mathrm{ord}(g)$ *(verified for all 25920 elements). Hence* any *order-3 symmetry of the substrate splits the matter register into eigenspaces of exactly $27 + 27 + 27$ — the three-generation decomposition is forced for every choice of triality element, not selected — and every order-9 element (5760 of them) refines it into nine 9-dimensional sub-generations. In the code's own characteristic the parameters survive:* $\mathrm{rank}_3\partial_0 = 39$*,* $\mathrm{rank}_3\partial_1 = 120$*,* $\dim H_1(\mathbb{F}_3) = 81$*, so* $[[240, 81, 4, 3]]_3$ *is exact over* $\mathbb{F}_3$*. The number of fermion generations is the defining-characteristic degeneracy of the protected register.* Witness: `bt863_generations_from_steinberg_vanishing.py`.

Theorem 10.3 (The dual-torsor Steinberg compiler)**.** *Let $H_{27} = 3_+^{1+2}$ be the canonical Heisenberg O_3 of the point parabolic and let $A_{27} = \mathbb{F}_3^3$ be the canonical translation O_3 of the Bell-line parabolic. Their 27-point and 27-line shells are regular torsors (BT858). On the protected register,*

$$\mathrm{St}{\downarrow_{H_{27}}} = 3\,\mathbb{C}[H_{27}], \qquad \mathrm{St}{\downarrow_{A_{27}}} = 3\,\mathbb{C}[A_{27}],$$

because both restricted characters are exactly $\{81^1, 0^{26}\}$. The statement survives in the code's own field and is constructive:

$$H_1(W(3,3);\mathbb{F}_3){\downarrow_{H_{27}}} \cong \mathbb{F}_3[H_{27}]^{\oplus 3}, \qquad H_1(W(3,3);\mathbb{F}_3){\downarrow_{A_{27}}} \cong \mathbb{F}_3[A_{27}]^{\oplus 3}.$$

For each torsor the verifier finds three cycle seeds whose 27 group translates add $27 + 27 + 27$ independent classes modulo the 120-dimensional boundary space. Thus the same logical memory has a flat program basis and a curved state basis: A_{27} is the commuting three-trit address compiler, while H_{27} is the noncommuting qutrit-phase compiler.

The center $Z(H_{27}) = \langle z \rangle \cong C_3$ selects a canonical triality. Over $\mathbb{C}$ its central-character blocks are

$$H_1 = H_{1,1} \oplus H_{1,\omega} \oplus H_{1,\omega^2}, \qquad \dim H_{1,j} = 27.$$

The trivial block contains the nine linear characters with multiplicity 3; the two nontrivial blocks contain the conjugate 3-dimensional Schrödinger representations with multiplicity 9. Over $\mathbb{F}_3$, where $x^3 - 1 = (x-1)^3$, the phases coalesce into a canonical unipotent flag. With $N = z - I$,

$$(\mathrm{rank}\, N, \mathrm{rank}\, N^2, \mathrm{rank}\, N^3) = (54, 27, 0),$$

so

$$0 \subset \ker N \subset \ker N^2 \subset H_1, \qquad \dim = (0, 27, 54, 81),$$

has three successive quotients of dimension 27. Equivalently, z acts by 27 Jordan blocks of length 3. Complex generations are phase sectors; native qutrit generations are the three layers of one depth-3 memory stack.

The freeness is canonical, but the displayed orbit bases are not: H_{27} and A_{27} are nonisomorphic, and a basis-to-basis matrix still requires a chosen point, line, shell origin, and cycle seeds. No canonical intertwiner is claimed. Witness: `bt865_dual_torsor_steinberg_compiler.py`.

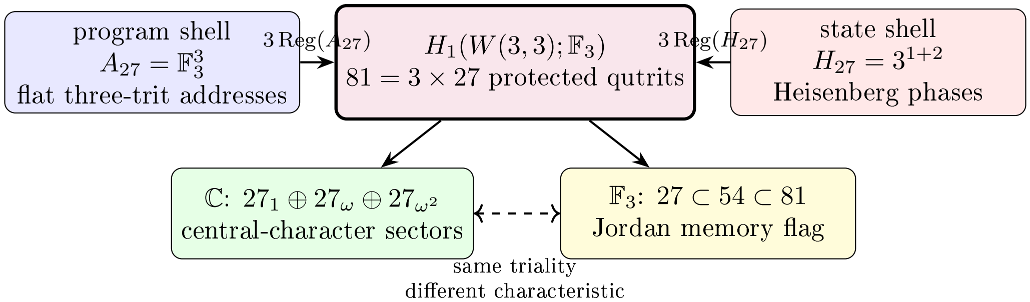


Figure 7: The dual-torsor compiler. Flat program translations and noncommutative state translations give two free rank-3 module coordinates on one Steinberg-protected register. The Heisenberg center reads as three phase sectors over $\mathbb{C}$ and as a three-step unipotent memory flag over $\mathbb{F}_3$.

Theorem 10.4 (Steinberg memory)**.** *The Levi-graph cycle space and the chart-overlap* 81*-sector are both irreducible and equal to the Steinberg representation of* $\mathrm{PSp}(4,3)$ *— the Solomon–Tits homology of the building, whose canonical basis is the* 81 *apartments through any chamber. Schur's lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow* 3*-subgroup the module is free of rank one: the protected memory is one free generator over the substrate's ternary symmetry.* Witnesses: `bt742`, `bt744`.

Corollary 10.5 (The oriented timetable spectrum)**.** *Let* $P = 3^3{:}S_4$ *be the index-*40 *line parabolic. The ordinary line permutation module and the oriented top-homology module are the two inductions of its two linear characters:*

$$\mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(1) = 1 \oplus 15 \oplus 24, \qquad H_2 = \mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(\mathrm{sign}_{S_4}) = 5_\omega \oplus 5_{\omega^2} \oplus 30.$$

The two 5-dimensional characters are complex conjugates over $\mathbb{Q}(\zeta_3)$*; the 30 is rational. Under the outer extension* $W(E_6) = U_4(2){:}2$*, the conjugate pair is the restriction of one irreducible 10-dimensional representation, whereas the 30 has two inequivalent extensions. Thus the oriented timetable header reads as a Weyl-visible chiral channel* 10 *plus a payload* $30^{\pm}$ *whose outer parity is not fixed by the projective module alone.*

The completed homology spectrum is

$$H_0 = 1, \qquad H_1 = \mathrm{St}_{81}, \qquad H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30,$$

and resolves the Euler charge representation by representation:

$$1 - 81 + (5 + 5 + 30) = -40 = -v.$$

BT867 resolves the tempting cache comparison negatively: neither 162*-cache branch is one of the two 5-dimensional sectors. The branches are isomorphic copies with identical character overlaps; the outer involution acts on their multiplicity label. The possible identification of the two 30-extensions with BT857's local gauge bit remains open.* Witness: `analysis/bt866_h2_oriented_irreducible_decomposition.py`.

Theorem 10.6 (The cache/transport extension boundary)**.** *Let* $H = 3^3{:}S_4$ *be the order-*648 *Bell-line parabolic. The two* 162*-slot BT859 cache branches have conjugate cyclic stabilizers and are therefore isomorphic transitive* H*-sets,*

$$\mathcal{C}_L \cong H/C_4 \cong \mathcal{C}_R.$$

Their permutation characters are equal. In particular, their scalar products with the restrictions of the three BT866 characters $(5_\omega, 5_{\omega^2}, 30)$ *are both* $(1,1,8)$*: each cache sees both conjugate 5-sectors equally. "Left" and "right" are copy labels, not internal spectral chiralities.*

The parabolic contains one conjugacy class of 81 *cyclic* C_4 *subgroups, and*

$$N_H(C_4) = D_8, \qquad H/C_4 \longrightarrow H/D_8, \qquad 162 \longrightarrow 81$$

is the canonical free two-sheet cache cover. Both copies cover this same stabilizer base. On their union GAP computes

$$C_{\mathrm{Sym}(324)}(H) \cong D_8, \qquad 324 = 81 \times 4 = 81 \times 2_{\mathrm{deck}} \times 2_{\mathrm{cache}},$$

with exactly 81 *commutant orbits of size four. Thus the four-state fiber carries the symmetry of a square, not merely a numerical pair of bits.*

This cache carrier is not the older ternary 162*-sector. On one cache fiber the deck swap and its two projectors are*

$$D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_\pm = \frac{I \pm D}{2}, \qquad \mathbb{F}_3^2 = \operatorname{im} P_+ \oplus \operatorname{im} P_-.$$

Since 2 *is invertible in* $\mathbb{F}_3$*, the cache splits; indeed* $(D-I)^2 = D - I$*. By contrast, packet transport uses*

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0, \qquad (I+N)^3 = I,$$

and tensors this indecomposable fiber with the 81*-register to obtain the non-split sequence* $0 \to 81 \to 162 \to 81 \to 0$*. No change of basis can identify the idempotent cache difference with the nilpotent transport shift. The architecture therefore separates a reversible* address/control plane *from a memory-bearing ternary* temporal data plane *by extension class, not by convention.* Witness: `analysis/bt867_cache_split_transport_nonsplit_boundary.py`.

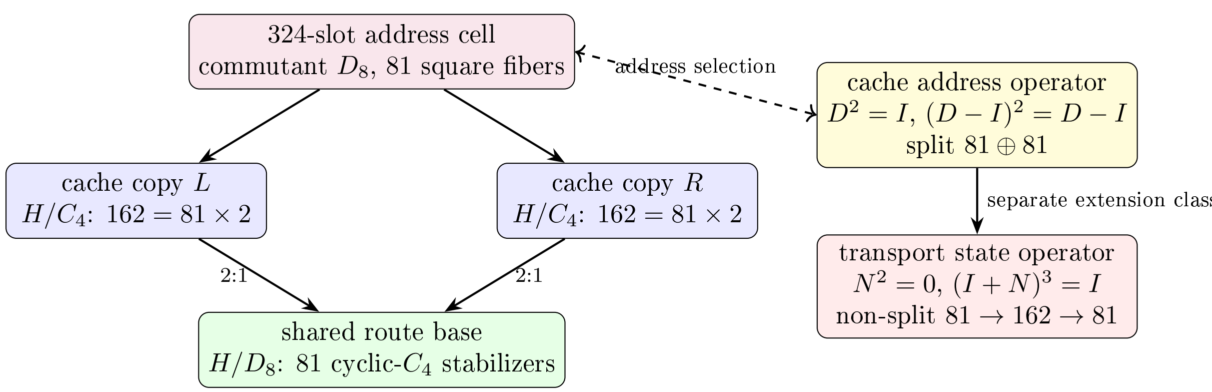


Figure 8: The local cache cell and the split/non-split boundary. The D_8 square fiber chooses a reversible address over one of 81 routes; the independent ternary nilpotent updates state while retaining temporal memory. Equal 162-dimensional counts do not imply equivalent modules.

The error layer is native: the $[[240, 81, 4, 3]]_3$ CSS code on the edge qutrits has all four parameters substrate-primitive; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web's unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

11 Time: three clocks and a quasicrystal

Theorem 11.1 (Clock trichotomy). *Of the program's three natural clocks, exactly one is internal:* $\mathbb{Z}_{12}$ *(the rectangle clock,* $12 \mid 51840$*) runs the selector and duo machinery, while* $\mathbb{Z}_7$ *(Császár) and* $\mathbb{Z}_{13}$ *(Singer) do not divide the group order and act as external references — both with decimal period* $6 = q!$*, so the digit expansion of* $1/7$ *is literally the internal clock's shadow modulo its duo bit.* Witnesses: `bt774`, `bt803`, `bt807`, `bt819`.

Theorem 11.2 (The Boerdijk–Coxeter time quasicrystal). *Driving the loop by the BC twist* $\theta = \arccos(-2/3)$ *per round trip yields a stroboscopic orbit that provably never repeats (Niven), with the Steinhaus three-distance signature at all substrate step counts and — the substrate's signature moment — exactly* two *gap lengths at* $n = 30 = h(E_8)$*, the BC ring length, where closure happens in* S^3 *(the* 600*-cell) rather than the phase circle (deficit* 0.0158*). This is the photonic sibling of the Fibonacci-drive dynamical topological phase.* Witness: `bt820`.

Theorem 11.3 (The arrow of time). *Without its two-trit feedforward record, the photon's self-configuration loop is the completely depolarizing channel with unique fixed point* I/q *— exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at* $\log_3 q^2 = 2$ *trits per cycle.* Witness: `bt823`.

12 Synchronization: beacons and schedules

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete K_7 interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3, 3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses:* `bt817`*,* `bt819`*.*

Theorem 12.1 (Schedule-overlap law). *Two distinct schedules share exactly* 1 *or exactly* $4 = \mu$ *contexts: each schedule has* 15 near *partners (overlap* 4*) and* 20 far *partners (overlap* 1*), the two relation classes of the double-six diagonal* $[1, 15, 20]$*, with the exact counting identity* $1 \cdot 20 + 4 \cdot 15 = 80 = 10 \times (9 - 1)$*. Operationally: a cheap timetable switch preserves* 4 *of* 10 *calibrated contexts and a far switch only* 1*, so a controller hopping among near partners re-calibrates* 6 *contexts per hop — the timetable library is itself a metric space whose geometry is the double-six relation.* Witness: `bt835_schedule_overlap_theorem.py`.

The Grünbaum–Coxeter cells. Each schedule also has an internal shape. The full schedule stabilizer S_6 is 2-homogeneous on the 10 contexts, but its icosahedral core A_5 — the 600-cell level of the platonic ladder, and *exactly* the automorphism group of the hemi-dodecahedron $\{5, 3\}_5$ — splits the $\binom{10}{2} = 45$ context pairs as $45 = 15 + 30$, and the 15-orbit graph on the 10 contexts is the **Petersen graph**: the edge skeleton of the hemi-dodecahedron, the cell of Coxeter's exceptional 57-cell. Dually, the hidden 6-set of the S_6 action realizes the hemi-icosahedron $\{3, 5\}_5$ — the cell of Grünbaum's 11-cell — with incidence count $(6, 15, 10)$: six hidden points, fifteen duads, ten triple-splits = the contexts themselves. The exceptional polytopes' groups do not embed in $\mathrm{Sp}(4, \mathbb{F}_3)$, but they are substrate arithmetic ($|\mathrm{PSL}(2, 11)| = 660 = k \cdot N_{\text{eff}}$, $|\mathrm{PSL}(2, 19)| = 3420 = k \cdot g \cdot 19$, Heawood genus $H(19) = 20 =$ the BC ring count, and $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$): the full 11-cell and 57-cell live elsewhere, yet their *cells* are carried by every measurement timetable of this machine, glued by the substrate's own group instead of $\mathrm{PSL}(2,p)$. *Witness:* `bt836_gc_hemicells_in_spreads.py`.

Theorem 12.2 (The library is a classical geometry). *The timetable library carries its own rank-3 geometry, and the hemicell structure fibers over it with perfect uniformity: (i) every skew line pair lies in exactly* 3 *schedules, so the* (schedule $\supset$ pair) *flag count is* $36 \times 45 = 1620$ *— the apartment count of the Tits building — and* 540×3*; (ii) the near-partner graph on the* 36 *schedules is strongly regular* $\mathrm{SRG}(36, 15, 6, 6)$*, the* other *classical rank-3 graph of* $U_4(2) \cong \mathrm{PSp}(4, \mathbb{F}_3)$*, so the control plane (timetables) and the data plane (*$W(3,3) = \mathrm{SRG}(40, 12, 2, 4)$*) are the two rank-3 geometries of one group; (iii) each schedule stabilizer contains exactly* 6 *icosahedral* A_5 *cores (*$216 = 6^3$ *in all, each marking its unique schedule via its* $[10, 30]$ *line orbits) giving* 6 *distinct Petersen splits; each internal pair is a Petersen edge under exactly* 2 *of the* 6 *cores, hence every skew pair of* $W(3,3)$ *has exactly* $6 = 3 \times 2$ *hemi-dodecahedral homes (*$3240 = 540 \times 6$ *Petersen flags).* Witness: `bt837_schedule_library_geometry.py`.

Theorem 12.3 (Two compasses, twelve Petersens, $4 \cdot K_{10}$). $\mathrm{PSp}(4, \mathbb{F}_3)$ *has exactly two conjugacy classes of* A_5*, both with normalizer* S_5 *and* 216 *conjugates, unfused by the outer automorphism, and* both fiber over the same 36 schedules*: every schedule stabilizer contains* $12 = 6 + 6$ *icosahedral cores —* 6 duad *cores (line signature* $[10, 30]$*, the Theorem-12.2(iii) cores) and* 6 pentad *cores (signature* $[5, 5, 10, 20]$*: transitive on the schedule, plus two distinguished* pentads *of* 5 *pairwise-disjoint lines outside it). The two* 216*-sets are non-isomorphic G-sets (rank* 10 *suborbits* $[1, 5, 10, 10, 20, 20, 20, 30, 40, 60]$ *vs* $[1, 5, 10, 10, 20, 20, 30, 30, 30, 60]$*); the Petersen flags form the coset geometry* $\mathrm{PSp}(4, \mathbb{F}_3)/D_4$ *(*3240 *flags, dihedral stabilizer). There are* 432 *distinct pentads in* two chiral orbits *of* 216 *(stabilizer order* 120*); every core pairs one LEFT with one RIGHT pentad, and each pentad serves exactly one core. Pentads are* maximal partial spreads*: no schedule contains one. The two pentads of one core interlock as* $K_{5,5}$ *minus a perfect matching — mutual disjointness is* forbidden *by the overlap law (it would create a* 0*-overlap schedule) — and the deleted matching consists of* 5 skew *pairs, i.e. five hypercube charts; over the* 216 *cores these matchings cover the* 540*-chart atlas* ***exactly twice*** *(*$1080 = 540 \times 2$*): the routing fabric is readable off the pentad compasses, five charts per needle. Finally,* all twelve *cores split the* 45 *internal pairs as Petersen* 15*-orbits, and the twelve Petersen edge-sets cover every pair exactly* $\mu = 4$ *times:*

$$4 \cdot K_{10} \;=\; 12 \textit{ Petersens}.$$

Schwenk's classical obstruction (no edge-partition of K_{10} *into* 3 *Petersen graphs) dissolves at the substrate multiplicity* μ*. The numerical coincidence* $3240 = \#\{\textit{triangles of } Q\}$ *is refuted at the G-set level (orbits* $[360, 2880]$ *vs transitive).* Witnesses: `bt843_icosahedral_compass.py`, `bt844_double_five_and_six_petersens.py`, `bt845_pentad_exchange_and_chart_double_cover.py`.

Theorem 12.4 (The amalgamation ladder). *The* 11*-cell and* 57*-cell are* universal *abstract polytopes: the unique ones with hemi-icosahedral facets and hemi-dodecahedral vertex figures (group* $L_2(11)$*, no proper quotients) and vice versa (*$L_2(19)$*). Their incidence is icosahedral-subgroup geometry: each of* $L_2(11)$*,* $L_2(19)$*,* $U_4(2)$ *has exactly two conjugacy classes of* A_5*, and in all three the distinguished incidence is* intersection in D_{10} *with valency* 6 *— the* 11*-cell's vertex–facet relation, the* 57*-cell's facet adjacency (the Perkel graph), and the substrate's compass incidence (*$216 + 216$*, bipartite* 6*-regular,* $216 \times 6 = 1296 = 6^4$ *incident pairs* $= 36$ *schedules* $\times\, 36$ *core pairs). The closures of one* *amalgam $A_5 *_{D_{10}} A_5$ (GAP):*

$$\langle A,B\rangle = \begin{cases} L_2(9)\cong A_6\ (360) & \textit{in } U_4(2)\textit{: inside exactly one schedule stabilizer,}\\ L_2(11)\ (660) & \textit{the } 11\textit{-cell,}\\ L_2(19)\ (3420) & \textit{the } 57\textit{-cell.}\end{cases}$$

One local icosahedral amalgam, a PSL(2)*-ladder of completions over* $9 = q^2$*,* $11 = k-1$ *(the Ihara prime),* 19 *(Heawood,* $H(19) = 20$*). Universality plus Lagrange (*$11, 19 \nmid 25920$*) shows the substrate carries the* complete local data *of both exceptional polytopes while their rank-*4 *amalgamation is arithmetically forbidden inside* $\mathrm{Sp}(4,\mathbb{F}_3)$*: the GC polytopes are the substrate's sibling completions, not subobjects. One step up the universal ladder (*$\{\{5,3\},\{3,5\}_5\}$*,* 10006920 *facets) the completion group is* $J_1 \times L_2(19)$ *— the first sporadic Janko group, whose involution centralizer* $2 \times A_5$ is *the compass datum. Finally, the compass Petersen* 15*-orbit carries exactly* 12 *pentagons splitting under the compass into two* 6*-orbits, each covering every edge exactly twice: two* chiral fully-faced *hemi-dodecahedra per needle, not merely skeletons. The dark sector closes the calculus: its* 190 *line pairs split* $[10, 30, 30, 30, 30, 60]$ *under the core into a perfect matching of* 10 dark charts*, two chiral* $5\times K_4$ *partitions into skew tetrads, the two dark dodecahedra, and a* 6*-regular meeting frame; the dark matching is* schedule-shadow pairing *(each dark line meets exactly* 4 *schedule lines; matched lines share their shadow), and in* K_5 *terms the ten shadows are exactly the ten "edge + opposite triangle" configurations — a canonical bijection dark chart* $\leftrightarrow$ K_5 *edge* $\leftrightarrow$ *schedule line. The timetable is stored three ways (lit-chart transversals,* K_5 *edges, dark shadows): built-in erasure redundancy. Each chiral tetrad's four distinguished edges form a* K_5 star*, the five tetrads of each partition marking the five lit charts exactly once:* K_5 *vertices are stored three ways (lit charts, left/right tetrads), edges two ways (schedule lines, dark charts) — every* A_5*-orbit structure of the pentad core except the dodecahedra and the meeting frame is a* K_5 *element — and those two fold in as well: each chiral dark dodecahedron is the* antipodal double cover *of the Petersen (Kneser) graph on the* 10 *dark charts, with deck transformation the shadow matching (antipode = shadow partner, verified), and the meeting frame is the doubled Johnson graph* $T(5)$*. The* K_5*-relation census is constant on every orbit (matching = same edge, tetrads and meeting = adjacent, dodecahedra = disjoint): the dark sector is the canonical* 2:1 *cover of the* K_5 *edge-world, with the Petersen graph appearing three ways in one needle (schedule skeleton, dark Kneser quotient, doubled chiral covers).*

Theorem 12.5 (The dark charts are the mirror bus)**.** *The* (core, dark chart) *slot space —* 216 *pentad cores* $\times 10$ *dark charts =* 2160 *slots, with every skew pair occurring as a dark chart in exactly* 4 *cores (*$2160 = 540 \times 4$*, the repair-atlas factorization) — is* ***G-set isomorphic to the*** 2160***-slot*** D_{12} ***mirror bus*** *of the middleware: the slot stabilizer has order* 12 *with the* D_{12} *profile* $\{1{:}1, 2{:}7, 3{:}2, 6{:}2\}$ *and is* conjugate in $\mathrm{PSp}(4,\mathbb{F}_3)$ *to the antipode-slot stabilizer (direct search over all* 25920 *elements). The multiplicity-*4 *agreement is canonical: a dark chart's transversal tetrad is its schedule shadow, so the* 4 *host cores' schedules all contain the pair's* 4 *transversals. Hence the compass layer* generates *the middleware — timetables from its lit side (reconstruction), the transport bus from its dark side — and the mirror bus acquires a hardware identification: one slot = one needle's dark chart.* Witnesses: `bt856_dark_mirror_bus.py`; further: `bt853_dark_orbit_zoo.py`, `bt854_dark_chart_transversal_duality.py`, `bt855_dark_sector_k5_complete.py`.

Witnesses: `bt848_universal_amalgam_ladder.py`, `.tmp/gap_bt848*.g`; cf. Hartley's classification (17 universal rank-4 locally projective polytopes, 441 quotients) and Hartley–Leemans on $\{5, 3, 5\}$.

The barren rung and the maniplex workaround. Relaxing polytopes to *maniplexes* (flag graphs without the intersection property; the tomotope middleware of this machine is a uniform 4-polytope whose monodromy $18432 = 96 \times 192$ fails that property, giving it *infinitely many* distinct minimal regular covers — the canonical rank-4 cover pathology) does not lift the obstruction: exhaustive search shows $U_4(2)$ admits *no* rank-4 regular maniplex with a 5 in its type, and the amalgam closure $A_6 = L_2(9)$ admits no regular maniplex at any rank — it is the *barren rung* of the PSL(2) ladder, whose populated rungs are exactly the 11-cell (q=11), the 57-cell (q=19), and Leemans–Toledo's degenerate single-facet $\{5, 5, 5\}$ family (our control search rediscovers all of them, including both exceptional polytopes). This is why the Grünbaum–Coxeter content of the substrate is *delocalized* over the 36 schedules — Petersen splits, chiral pentads, chart covers, dark dodecahedra — rather than crystallized in one flag object, and why the machine's runtime uses the tomotope: it is the flag structure that survives the obstruction. The tomotope's symmetry type is now exact: $|\mathrm{Aut}| = 96$ (independent centralizer computation on the 192 flags), two flag orbits $[96, 96]$, **class** $2_{\{0,1,2\}}$ — only the cell-color crosses orbits, and the orbits are precisely "flag in a tetrahedron" vs "flag in a hemioctahedron": the middleware *alternates spherical and projective cells*, the exact situation the classical "locally X" taxonomy cannot name, and the setting of Mochán's 2-orbit polytopality theory. Moreover the tomotope is a **Cayley maniplex over** $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ (vertex action = full S_4 with a regular Klein subgroup; edge and face actions faithful and transitive): the middleware's vertex layer is a free $\mathbb{F}_2^2$-torsor, so register XOR-addressing extends from the chart atlas into the runtime with no added structure. Reading Hartley's full classification (seventeen universal locally projective rank-4 polytopes) against the substrate adds three facts: Aut(**tomotope**) **is itself a universal polytope group** — isomorphic to $\Gamma(\{\{4, 3\}_3, \{3, 4\}_3\})$, the quotient-free doubly-projective $\{4, 3, 4\}$ universal built from hemicubes and hemicrosses (the tomotope's own projective cell type), with the case-10 group $= C_2 \times \mathrm{Aut}$ of order 192 = the flag count; the mixed $\{3, 5, 3\}$ amalgam (icosahedral facets, hemi-dodecahedral vertex figures) *does not exist* — the construction closes into the 11-cell — while the dodecahedral mix is the J_1 giant; and the chart–compass universal $\{\{4, 3\}, \{3, 5\}_5\} = 2^{\mathcal{I}}$ (cube facets = Q_3 charts, hemi-icosahedral vertex figures = compass cells, vertices = $\mathbb{F}_2^6$ on the hidden 6-set) has order $3840 \nmid 25920$: blocked like the rest of the ladder. *Witnesses:* ***bt849_ maniplex_ barren_ rung.py, bt850_ tomotope_ two_ orbit_ class.py, bt851_ tomotope_ cayley_ test.py, bt852_ seventeen_ universals.py, .tmp/gap_ bt849*.g, .tmp/gap_ bt852*.g.***

Reconstruction, the chart K_5, and the dark dodecahedra. The pentad core's full signature $[5, 5, 10, 20]$ (lines), $[20, 20]$ (points) is rigid: both chiral pentads cover the *same* 20-point orbit (lit/dark halves of the substrate), the five matching charts' common-transversal bundles overlap to exactly the 10 schedule lines — **a pentad pair generates its timetable** — and the line $\to$ chart-pair map is a bijection onto the $\binom{5}{2} = 10$ edges of K_5: the schedule *is* a K_5 on its own charts (4 transversals per chart, 2 charts per line). The 20 dark lines complete the $\{5, 3\}$ family: their 190 pairs split $[10, 30, 30, 30, 30, 60]$ under the core, and exactly two of the 30-orbits are **dodecahedron skeletons** (3-regular, girth 5, diameter 5) — a chiral pair of genuine dodecahedra beneath every compass needle, one level under the hemi-dodecahedral Petersen splits of the schedule itself. *Witnesses:* ***bt846_pentad_core_anatomy.py, bt847_dark_dodecahedron.py.***

Theorem 12.6 (BT839 GC operation Euler/flag audit)**.** *Treating the full Grünbaum–Coxeter note as a data table gives a second invariant. The base tables for the* 11*-cell,* 57*-cell, tomotope partial a, and tomotope partial b are all Euler-neutral:*

$$V - E + F - C = 0.$$

But every Wythoff operation table has a fixed family charge:

$$\chi_{\mathrm{op}}(11\text{-}cell) = -11, \qquad \chi_{\mathrm{op}}(57\text{-}cell) = -57, \qquad \chi_{\mathrm{op}}(tomotope) = -4.$$

The omnitruncated rows are full-flag carriers:

$$660 = |\mathrm{PSL}(2,11)| = 11 \cdot A_5 = kN_{\mathrm{eff}},$$

$$3420 = |\mathrm{PSL}(2,19)| = 57 \cdot A_5 = k \cdot g \cdot 19,$$

and the tomotope has 96 *half-flags, whose double is the verified* 192*-flag BT814 packet. The* 57*-cell also locks to the W33 timetable library: BT837 gives* 3240 *Petersen homes, and*

$$3240 + k \cdot g = 3240 + 180 = 3420.$$

Thus one $k \cdot g$ *sentinel sheet completes the W33 Petersen-home carrier to the* 57*-cell flag count.*

The regular-polychoron alignment supported by the cell/symmetry data is

$$11\text{-}cell \to 600\text{-}cell, \qquad tomotope \to 24\text{-}cell, \qquad 57\text{-}cell \to 120\text{-}cell.$$

The $57 \to 120$ *link is direct at the dodecahedral cell level; the* $11 \to 600$ *link is weaker but real, through hemi-icosahedral local structure and the quotient* $2I \to A_5$ *rather than through a direct tetrahedral-cell match. The swapped* 11/57 *orientation is recorded as a lower-scoring dual/vertex-figure hook, not discarded.* Witness: `bt839_gc_operation_euler_flag_audit.py`.

Theorem 12.7 (BT840–BT842 carrier closure)**.** *The three unresolved carriers in the Grünbaum–Coxeter paragraph are now explicit objects.*

(i) The* 180 *sentinel sheet. *The verified Clifford L/R boundary has* 36 *cells arranged as a* 6×6 *rook grid. Its zero-overlap relation is not a numerical afterthought: it is the union of the* 6 *row fibres and* 6 *column fibres, each fibre carrying the duads of a six-set. Hence*

$$12 \cdot \binom{6}{2} = 12 \cdot 15 = k \cdot g = 180.$$

This is exactly the sheet missing from BT837:

$$3240 + 180 = 3420 = |\mathrm{PSL}(2,19)|.$$

Equivalently, the timetable library gives 216 *Petersen cores (*$216 \cdot 15 = 3240$*), and the rook boundary supplies the missing* 12 *six-set fibres (*$12 \cdot 15 = 180$*).*

(ii) The* 660 *carrier. *At an apex cell* (r, c) *of the same* 6×6 *grid, remove the incident row and column. The apex plus the five nonincident rows and five nonincident columns gives*

$$11 = 1 + 5 + 5.$$

Crossing these eleven labels with the verified 60*-element Clifford* A_5 *selector gives*

$$11 \cdot A_5 = 11 \cdot 60 = 660 = kN_{\mathrm{eff}}.$$

This is a carrier theorem only: it does not assert an internal $\mathrm{PSL}(2,11)$ *action on* $W(3,3)$*.*

(iii) The* 96 *tomotope half-flags. *In the* D_4*-root model of the 24-cell, every edge lies in a unique central hexagon plane. The two endpoint axes determine the third, missing axis in that* *plane, so each edge maps to a Reye incidence (missing axis, hexagon plane). Each of the* 48 *Reye incidences has exactly two edge lifts:*

$$96 = 2 \cdot 48.$$

Thus the tomotope omnitruncated half-flag count is literally the 24*-cell edge lift of the common tomotope/Reye spine.*

The hexagon clue supplies the conceptual bridge to the 11*-cell. Each central* 24*-cell hexagon is a six-root cycle* C_6*. Completing that cycle to a full* K_6 *gives* 15 *duads, exactly the hemi-icosahedral skeleton count of the* 11*-cell cell. Across the* 16 *central hexagons,*

$$16\binom{6}{2} = 16 \cdot 15 = 240,$$

the W33 edge count and the E_8 *root count. The tested dot profile of these* 240 *duad slots is*

$$96_{\langle x,y\rangle=1} \; + \; 96_{\langle x,y\rangle=-1} \; + \; 48_{\langle x,y\rangle=-2},$$

namely live 24*-cell edges, mirror pairs, and repeated opposite-axis Reye incidences.* Witnesses: BT840, BT841, and BT842; executable scripts are in `analysis/`.

13 Build sheet and falsifiable witnesses

Components (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

Witnesses (kill criteria, all parameter-free):

witness	predicted value	substrate form
trace–Choi visibility of F_3	$1/3$	$1/q$
trace–Choi visibility of X, Z	0	—
Kochen–Specker classical budget	$36/40$	$(q!)^2/v$
beacon-mesh pair visibility	$1/3$ (all 21 pairs)	$1/q$
Werner separability threshold	$3/4$	q/μ
BC-drive gap census at n=30	2 gap lengths	$h(E_8)$ ring
Hashimoto transport phases (one tick)	$\arctan\sqrt{10},\ \pi - \arctan(\sqrt{7}/2)$	Φ_4, Φ_6; $\|u\|^2 = 11$
key-agreement rate	$13/40$	Φ_3/v
D_{12} mirror-slot stabilizer	order 12	projective bus
runtime factorization	$24 \cdot 45 \cdot 48$	$\|\mathrm{Sp}(4, \mathbb{F}_3)\|$
one recursive route digit	≤ 8 reversible hops	$3 + 5$
level-n leaves	40^n	v^n
level-n $W(3,3)$ instances	$(40^n - 1)/39$	geometric tower
durable commit clock	$4(7^g - 1)$ ticks	tomotope/Csaszar
compiled stress route	$47 < 48$ moves	BT828 level six
cover-lifted packet capacity	$48k^3$ slots	BT832 storage gauge
Bell–compass prefix router	$648 + 324 + 162 + 162$	code $0, 10, 110, 111$
sentinel-aware stress energy	$4 \to 2$	BT833 reroute
full route sync criterion	$2n \mid 7^n - 1$	BT834 guard band
tomotope Wythoff runtime ladder	$4, 12, 24, 48, 96$	BT838 packet growth
GC operation Euler charges	$-11, -57, -4$	BT839 table audit
sentinel projector trace	15	g sector
first full-route desync	$n = 5$ remainder 24	f guard band
tomotope cover ABI	48 blocks / 192 flags	cover-invariant

A measured deviation in any row falsifies the corresponding layer.

14 Implications, corrections, and the ethos

This section states what the architecture means — for physics, for computing, for networking, and beyond the laboratory. Every claim below is either (i) a theorem with an executable witness in this paper's ledger, or (ii) an implication of such a theorem, flagged as such. Nothing here floats free of the verification chain.

14.1 Physics

The vacuum is a gauge choice among five. The decomposition $40 = 1 + 12 + 27$ (self, gauge, matter) that organizes the entire Standard-Model derivation is *one of five* maximal decompositions, indexed by the Schläfli inventory of the cubic surface: 27 registers, 36 schedules, $40 + 40$ building parabolics, 45 polar pairs. The five vacua are connected by an explicit 5×5 double-coset transition matrix. Physically: what looks like "the" vacuum is a parabolic gauge choice, and the theory knows its own alternative phases and the exact combinatorial cost of moving between them. No continuum field theory has this property; here it is finite, exact, and enumerated.

Matter is magic, with three octane grades. The machine's non-classical resource (magic, in the resource-theoretic sense) coincides *exactly* with its matter sector: the 36 entangled Witting rays are the matter shell, and the 27 matter contexts stratify $[1, 8, 12, 6]$ — the cube's cell census. The three magic grades — deep $(2 + \sqrt{3})/6$ on $8 = 2^q$ rays, mid $(5 + 2\sqrt{3})/12$ on $24 = f$, shallow $3/4 = q/\mu$ (the Werner threshold) on $4 = \mu$ — mean that "what matter is" and "what fuels quantum advantage" are one census. A physicist reads this as a structural identification of the fermionic sector with the non-stabilizer sector; an engineer reads the same integers as a fuel gauge. Both readings are forced by the same double cosets.

Three generations, matter-blind and gauge-visible. Because the matter register is the Steinberg module (Corollary, §10) and the Steinberg character vanishes on every 3-singular element, *any* order-3 symmetry of the substrate splits matter into exactly $27+27+27$: the three generations carry identical matter/Steinberg structure — they are indistinguishable to the protected register, which is why generations share gauge charges. Yet the four order-3 conjugacy classes *are* distinguished in the gauge (40-point) sector: the transvections grade it $22 + 9 + 9$ and fix a whole $\mathrm{PG}(2, 3)$ plane of $13 = \Phi_3$ points, while the regular classes grade it $16+12+12$ and fix only $\mu = 4$ points. The *physical* texture triality (Pillar 68's Yukawa map R) is identified exactly: it is the long-root transvection (a 40-class element = the centre of the point-parabolic's Heisenberg radical), fixing the 13-point gauge perp-plane $p_0^\perp$ pointwise and acting on the 27-point matter shell with 9 free orbits of 3 — the canonical "fix the gauge, stir the matter" generator. Its eigengrading of $\mathbb{C}[27]$ is $9 \oplus 9 \oplus 9$, and $\mathbb{Z}_3$ Clebsch–Gordan then *forces* the Pillar-68 Yukawa selection rule $T[a, b, v] = 0$ unless the three grades sum to 0 mod 3 (exactly 9 of 27 grade-triples survive); the matter coupling lives on the 36 within-shell tritangent triples (Pillar 69's Heisenberg-twisted triads, $45 = 9 + 36$). And the gauge side closes it: the centralizer $C(R) = \mathrm{Stab}(p_0)$ of order 648 (the gauge group is what commutes with generation-cycling) acts on the $12 = k$ gauge neighbours with permutation rank 3, decomposing the gauge module as

$$\mathbb{C}[12] = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \oplus SU(2) \oplus SU(3),$$

the Standard-Model gauge group adjoint (the **1+3** from the A_4 action on the four lines through p_0, the **8** the within-line gluon octet). So R fixes the gauge sector ($12 = 8+3+1$, generation-blind) and grades the matter sector ($27 = 9+9+9$, three generations): the gauge group and the generations are the two eigen-data of one long-root transvection. The gauge sector even carries a parity statement: $C(R)$ acts on the four lines through p_0 as A_4 inside PSp, but as the full S_4 in $\mathrm{PGSp}(4,\mathbb{F}_3) = W(E_6)$ — the odd (parity-violating) 4-line permutations are *exactly* the anti-symplectic W/Q-duality coset, so gauge parity is the duality/chirality $\mathbb{Z}_2$ and lives only in the full $W(E_6)$. The flavor sector even has its charge conjugation: $N_{W(E_6)}(\langle R\rangle)/C(R) = \mathbb{Z}_2$, so there is an element C with $CRC^{-1} = R^{-1}$ that fixes generation grade-0 and swaps grade-1 $\leftrightarrow$ grade-2 — the discrete shadow of CP, conjugating the generation $\mathbb{Z}_3$ exactly as the temporal Bell qutrit's CPT conjugates $\omega \mapsto \bar{\omega}$. The full Standard-Model discrete flavor/parity data — gauge group $1 \oplus 3 \oplus 8$, three generations $\mathbb{Z}_3$, generation C, matter chirality, gauge parity — is the normalizer geometry of one long-root transvection. The grading R and its conjugation C together generate the **flavor group** S_3 ($\langle R, C\rangle$, order 6, non-abelian — the minimal non-abelian flavor symmetry of BSM model-building), under which the matter shell decomposes as $\mathbb{C}[27] = 6 \cdot \mathbf{1} \oplus 3 \cdot \mathbf{1}' \oplus 9 \cdot \mathbf{2}$, the standard doublet appearing $q^2 = 9$ times. And the flavor–gauge relationship is exact: R is central in the gauge group $C(R)$, with $Z(C(R)) = \langle R\rangle \cong \mathbb{Z}_3$ — **the three generations are the centre of the gauge group**, acting trivially on the adjoint exactly as the $\mathbb{Z}_3$ centre of colour $SU(3)$ acts on the gluon octet (so the $SU(3)$ centre *is* the generation $\mathbb{Z}_3$). Globally there are 40 such local gauge groups — one per point, a single conjugacy class (homogeneous spacetime) — and the 40 generation-centred long-root $\mathbb{Z}_3$'s generate all of $\mathrm{PSp}(4,\mathbb{F}_3)$: **the 40 points of $W(3,3)$ are the space of local gauge groups**, each $SU(3)\times SU(2)\times U(1)$ with its generation centre, and the global symmetry is the closure of the local generation symmetries. The connection between adjacent local gauge groups is exact: for collinear points the generation centres commute ($\langle R_p, R_{p'}\rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$), so the gauge connection is **flat along the 240 edges**; for non-collinear points $\langle R_p, R_{p'}\rangle = SL(2,3) = 2T$ (the binary tetrahedral / 24-cell group), so it is **curved across the 27-per-point matter shell**. Gauge curvature lives exactly on the matter graph Q — flat = collinearity, curved = non-collinearity = matter, with holonomy the 24-cell group. The curvature 2-form $F(p,q) = [R_p, R_q]$ makes this quantitative: it vanishes on collinear (causal) directions and is an *order-4 quaternion unit* of $2T$ (one of $\pm i, \pm j, \pm k$, with $F^2 = -I$ the centre) on every matter pair — an **$\mathbf{su}(2)$-valued field strength supported exactly on the matter graph Q**. Flat causality, quaternionic curvature on matter. The integrated flux (Wilson loop $R_aR_bR_c$) confirms it: all 160 collinear triangles carry order-3 abelian flux (the flat $\mathbb{Z}_3$ line grading), while the 3240 matter triangles of Q carry non-abelian flux of orders $\{2, 4, 6, 12\}$ (counts $180, 180, 1440, 1440$) up to $12 = k$ — gauge energy lives on the matter graph. This is the Standard-Model pattern as a representation- theoretic theorem: generations are identical in gauge charge (matter-blind) and differ only through gauge-sector–visible couplings (the Yukawa/mass structure), here the eigenvalue degeneracy of an order-3 symmetry rather than a modeling input. *Witnesses: `bt863_generations_from_steinberg_vanishing.py`, `bt864_triality_census.py`.* Order-6 symmetries see *both* internal gradings at once: an order-6 element factors as g^2 (generation, order 3) times g^3 (chirality, order 2 — the sign-twisted carrier of H_2), and since χ_{St} vanishes on every 3-singular power, its $\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$ eigengrading factors cleanly: each chirality half $(81 \pm \chi(g^3))/2$ carries 3 equal generations. For the dominant involution type ($\chi(g^3) = q^2$) the chirality split is $\mathbf{45 + 36}$ — the substrate's own tritangent and double-six counts — so the left/right matter split is tied to Schläfli geometry; the other type gives $39 + 42$. Generation $\times$ chirality $= 3 \times 2 = 6 = q!$. The dominant chirality involution is identified exactly: $\mathrm{PSp}(4,\mathbb{F}_3)$ has just two involution classes, and the 45-class (eight fixed points carrying the cube Q_3, centralizer $2T \times 2T$ of order 576) is the *central involution of a hyperbolic polar pair* $L \leftrightarrow L^\perp$ — so the left/right split of matter, $45 + 36$, is sized by the substrate's own tritangent-plane and double-six counts, and the matter chirality $\mathbb{Z}_2$ *is* the 24-cell polar-pair swap. *Witnesses:* `bt868_joint_generation_chirality_grading.py`, `bt869_involution_chirality_classes.py`.

Time has three hands and an irrational escapement. The internal $\mathbb{Z}_{12}$ rectangle clock, the external $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) references, and the Boerdijk–Coxeter drive $\arccos(-2/3)$ together realize a discrete time quasicrystal: a drive that never repeats yet has exactly bounded gap structure (two gap lengths at $n = 30 = h(E_8)$). The arrow of time appears as the unique self-loop fixed point I/q with its two-trit forgetting cost: irreversibility is priced, not postulated. Implication: any laboratory system realizing this architecture is also a clock whose long-term stability is protected by a number-theoretic irrationality rather than by feedback.

The exceptional polytopes are the theory's siblings, not its parts. The Grünbaum–Coxeter arc (Theorems 12.2–12.4) shows the substrate carries the complete local geometry of the 11-cell and 57-cell — hemi-icosahedra and hemi-dodecahedra in every schedule, with exact incidence — while universality plus Lagrange forbid their global crystallization inside $\mathrm{Sp}(4, \mathbb{F}_3)$. The same local amalgam $A_5 *_{D_{10}} A_5$ closes as $L_2(9)$ (the substrate's schedule core), $L_2(11)$ (the 11-cell), $L_2(19)$ (the 57-cell), with $J_1 \times L_2(19)$ — a sporadic group — atop the universal ladder. The physical moral is delocalization: the substrate *cannot* host a "GC condensate," so the icosahedral order is necessarily spread over the 36 schedules as compass structure. This is the same mathematical mechanism by which frustrated phases in condensed matter refuse local order parameters — realized here exactly, with the obstruction an integer $(11, 19 \nmid 25920)$.

Spectral ghosts. The primes that refuse to divide the group still govern its analysis: $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$, 19 enters through the Heawood genus $H(19) = 20 =$ the BC ring count, and the chart-web eigenvalue fields are $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ ($\Theta = 10$, $\Phi_{12} = 73$, both parameter-closure constants). A spectroscopist's summary: the machine's transport spectrum speaks the tier-ladder dialect of the physics paper, and the forbidden primes appear as poles of the zeta function rather than as symmetries.

Gravity counts the matter register. The discrete gravitational bridge is the Matrix-Tree action $S = \frac{M_P^2}{2} \ln \tau(G)$, τ the spanning-tree count, and for the substrate it is exact: the Laplacian spectrum $\{0, 10^{24}, 16^{15}\}$ gives

$$\tau(W(3,3)) = \frac{10^{24} \cdot 16^{15}}{40} = 2^{81} \cdot 5^{23},$$

where the exponent of 2 is $81 = q^4 =$ the matter-sector (Steinberg) dimension, the base $5 = F_5$ is the tier-ladder prime ($r = 27/80$), and the exponent $23 = f - 1$. The complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ gives $\tau(Q) = 2^{66} \cdot 3^{39} \cdot 5^{23}$ with 3-exponent $39 = q\Phi_3 =$ the gauge-sector dimension. So the discrete-gravity partition function carries the Standard-Model sector sizes as its prime-exponent charges — matter (q^4) in the power of 2, gauge ($q\Phi_3$) in the power of 3, cosmological (F_5) shared — and the entropy of the spanning-tree ensemble *counts the registers.* This is not a separate fact from the transport spectrum: the Ihara zeta $\zeta^{-1}(u) = (1-u^2)^{200}(1-u)(1-11u)(1-2u+11u^2)^{24}(1+4u+11u^2)^{15}$ (verified directly by the 480×480 Hashimoto operator) has its non-trivial zeros on the graph-RH circle $|u| = 1/\sqrt{11}$ (the transport sector, $p_{\mathrm{Ih}} = k - 1 = 11$) and its Perron zero at $u = 1$, where the Bass matrix $I - A + 11I = 12I - A$ *is* the Laplacian, so the Matrix-Tree leading coefficient $10^{24}16^{15} = 2^{84}5^{24} = v \cdot \tau$ recovers exactly the gravity count. Transport and gravity are the off-critical and critical behavior of one zeta: the prime 11 is the transport critical norm, the matter dimension $q^4 = 81$ is the gravity entropy. The substrate is in fact a *pair* of zetas: the complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ has its own Ihara prime $26 = 2\Phi_3$ (Hashimoto Perron, all complex eigenvalues on $|u| = 1/\sqrt{26}$) and its Perron-point gravity $\tau(Q) = 2^{66}3^{39}5^{23}$ puts the *gauge* dimension $39 = q\Phi_3$ where $W(3, 3)$ puts the matter dimension — complementation (collinearity $\leftrightarrow$ non-collinearity) swaps the matter and gauge entropies, with the cosmological charge 5^{23} shared. The non-backtracking walk census confirms the dynamics: $N_3 = \mu|E| = 960$, $N_5 = |E|\, q^q\, n_{\text{even}} = 181440$, $N_n \sim 11^n$. *Witnesses:* *bt870_spanning_tree_gravity.py*, *bt872_ihara_zeta_spanning_bridge.py*, *bt873_dual_zeta_walk_census.py*.

Falsifiability is retail, not wholesale. Each physics reading above is pinned to a tabletop witness with an exact rational value: trace–Choi visibility $1/3 = 1/q$ exactly (not approximately); Kochen–Specker budget $36/40 = (q!)^2/v$ exactly, with deficit μ; contextual fraction $1/10 = 1/\Phi_4$; Werner threshold $3/4 = q/\mu$; two gap lengths — not three — at drive step 30. A single wrong integer kills the corresponding layer. This is a stronger falsifiability posture than parameter-fitting physics can offer, because there are no parameters to refit.

14.2 Computing

What the machine is. One photon carries a 4-dimensional path–polarization register (states) and a 9-dimensional past–future time-bin register (operators), realizing the same $W(3, 3)$ geometry twice; the operator–operand duality makes program and data literally the same incidence object. Universality is a two-ingredient theorem: three passive optical elements (tritter, phase plate, EOM) generate the full 51840-element Clifford symplectic closure — *exactly*, not approximately — and the matter shell supplies the magic states that promote Clifford to universal. There is no cryostat, no vacuum chamber, no trapped ion: the exotic resource is geometry.

What it costs. The build sheet is a heralded single-photon source, one polarizing beam splitter, one symmetric tritter, a three-rail delay ladder, one electro-optic modulator, one recirculation loop with a fixed irrational rotation, and single-photon detectors. Every component is catalog optics. The machine's benchmarks are integer-valued: a laboratory can certify the Clifford layer by counting to 51840 and certify the fuel by hitting 36/40. Self-testing is native because the specification is arithmetic.

The software stack exists and is finite. Programs are braid words (with $\sigma^5 = Z$ exact); the instruction set is the BT828 compiler ABI; the operating system is the timetable library (36 schedules; overlap law 1-or-4 with switching cost 6); synchronization is the beacon-heptad mesh (2880 frames, themselves forming an exchange network); memory protection is the Steinberg sector ($81 = q^4$, a single irreducible — protected by Schur's lemma, the strongest no-leakage statement representation theory can make). The state/program compiler makes that register directly addressable: $\mathbb{F}_3^3$ supplies three commuting copies of the 27-slot program shell, H_{27} supplies three noncommuting copies of the state shell, and its center supplies the native generation flag $27 \subset 54 \subset 81$. The oriented timetable header above that memory splits as $5_\omega + 5_{\omega^2} + 30$: the full Weyl runtime fuses the handed pair into a 10-channel and leaves one 30-channel with an outer parity choice. Cache lookup is a distinct split layer: two copies of H/C_4 share an 81-route H/D_8 base, and their 324 slots form 81 four-state D_8 address fibers. The non-split ternary transport operator then updates the selected state; address choice and temporal memory cannot alias because their minimal polynomials differ. The watchdog is the BC loop. Each layer is a theorem; the stack compiles because double cosets compose.

Redundancy is built in, not bolted on. The newest results sharpen the fault story: each pentad core stores its timetable *three independent ways* (lit-chart transversals, the K_5 edge labeling, dark-sector shadows), every skew pair has six Petersen homes and two matching covers, and the dark sector carries its own chart matching and two chiral tetrad partitions (Theorem 12.4 block). Erasure of any single description leaves the schedule reconstructible — redundancy appears as a counting identity, not as an engineering allowance. The middleware's two packet phases (tetrahedral/hemioctahedral, the tomotope's $2_{\{0,1,2\}}$ class) confine phase changes to cell hops, which is where the commit boundaries already sit, and the runtime's vertex layer is a free $\mathbb{F}_2^2$-torsor (the Cayley structure), so address arithmetic is XOR end to end.

What it is not. No claim is made here of asymptotic speedups over error-corrected gate-model machines, of fault-tolerance thresholds, or of scalability beyond the fractal substitution rule (40^n leaves, diameter $8n$, commit clock $T(n) = 4(7^n - 1)$) — those are stated as exact combinatorial scaling laws, and their physical realization at depth $n > 2$ is an open engineering problem with its first hard boundary already computed (the $n = 5$ desynchronization with remainder $24 = f$). The honest claim is narrower and stranger: a complete, finite, self-verifying computational architecture exists inside one photon's worth of quantum optics, with zero adjustable parameters.

14.3 Networking

The network is the computer, twice over. The thesis is not rhetorical: the atlas's transport groupoid and the register's gate groupoid are the same groupoid, so moving a packet and applying a gate are one operation. The new library-geometry theorem (Theorem 12.2) doubles the statement at the control plane: the 36 timetables form $\mathrm{SRG}(36, 15, 6, 6)$, the *other* rank-3 geometry of the same group, so the control plane and the data plane ($\mathrm{SRG}(40, 12, 2, 4)$) are the two classical geometries of one symmetry. Configuration management inherits strongly-regular guarantees: any two timetables, near or far, have exactly six common near-neighbors — worst-case reconfiguration paths are bounded by the geometry itself.

Routing, addressing, switching. Local routing is e-cube XOR on Q_3 charts (540 of them; the atlas is glued along the 1620 apartments of the Tits building, diameter 5). Addresses are Gray-coded $\mathbb{F}_2^3$ words; the antipode map is bit complement. Timetable switching obeys the overlap law: cheap hops preserve $4 = \mu$ of 10 calibrated contexts, the switch graph is the $U_4(2)$ rank-3 geometry, and every skew pair lies in exactly 3 schedules, 6 Petersen homes, and 2 pentad-matching covers — the router's spare-path inventory is an exact census, not a provisioning estimate.

Synchronization without a master. The beacon heptads (2880 complete K_7 interference meshes, all pairwise overlaps 1/3) are reference frames that form their own exchange network — the network's clock distribution is itself a network, with no distinguished master node. External discipline comes from the $\mathbb{Z}_7/\mathbb{Z}_{13}$ clocks (both decimal period 6) and the BC quasicrystal drive; internal phase order from C_{12}. A distributed-systems reader will recognize the structure: consensus by geometry, where the "leader election" problem dissolves because the frame family is a single transitive orbit.

Security posture (implication, flagged). Three structural facts have security readings, stated here as directions rather than theorems. (i) Contextuality is tamper-evidence: the KS budget 36/40 is exact, so any channel intervention that classicalizes measurement statistics moves an integer and is detectable in principle. (ii) Gate teleportation stores a unitary in the photon's self-entanglement and applies it by Bell projection against its own past — a store-and-forward primitive for quantum repeaters that never exposes the gate classically. (iii) The timetable redundancy (three independent reconstructions) is an erasure-code posture against denial: losing a description class does not lose the schedule. Turning these into security proofs requires adversary models this paper does not supply.

Scaling. The fractal holonet substitutes a full 40-node substrate for each leaf: 40^n leaves, $(40^n - 1)/39$ substrate instances, reversible diameter $\leq 8n = 8\log_{40} N$, durable commit clock $T(n) = 4(7^n - 1)$, with the first guard band an arithmetic necessity at $n = 5$. Because every child inherits the same finite protocol, scaling adds no new control logic — the protocol stack is depth-invariant. This is the architectural property that datacenter fabrics approximate statistically and this geometry has exactly.

14.4 The physics-to-architecture dictionary

The dictionary is the deepest implication: one set of integers reads simultaneously as physics and as engineering, because both readings are projections of the same incidence theorems.

object	physics reading	architecture reading
$1 + 12 + 27$ split	Bell line + meeting + disjoint	pole / I-O ports / fuel+store
240 edges	interaction qutrits	bandwidth = memory
$81 = q^4$ Steinberg	protected sector (Schur)	logical register
dual O_3 restrictions	Heisenberg state phases	state/program compiler
$H_2 = 5 + \bar{5} + 30$	oriented context chirality	timetable header $10 + 30^{\pm}$
$324 = 81 \times 4$, commutant D_8	split cache multiplicity	four-state address fiber
$0 \to 81 \to 162 \to 81 \to 0$	ternary temporal extension	non-split state update
36 spreads	measurement bases	timetable OS
$\mathrm{SRG}(36, 15, 6, 6)$	basis-overlap law	control-plane fabric
five vacua	phases of the theory	boot modes
magic strata 8+24+4	fermion shell grading	fuel octanes
540 skew pairs	two-fold null systems	hypercube charts
1620 apartments	Weyl frames	inter-chart trunks
2880 heptads	equiangular frames	sync beacons
BC drive $\arccos(-2/3)$	quasicrystal time	watchdog clock
compasses (216+216)	icosahedral order, delocalized	needle registers
pentads (chiral 216+216)	maximal partial spreads	F_5 chart caches
dark sector zoo	hidden-half structure	erasure store
tomotope ($2_{\{0,1,2\}}$, Cayley/V_4)	sphere/projective alternation	packet middleware
11-cell, 57-cell, J_1	universal completions (blocked)	the covers the hardware refuses

The first row carries the deepest identification, drawn from the companion paper *Universal Quantum Computation Through a Single Photon*: the split $1 + 12 + 27$ is not merely the local SRG anatomy of a point — it is the **Bell-line shell** of the photon's own self-entanglement seed. The temporal Bell qutrit $|\Omega\rangle$ stabilizes one line of $W(3, 3)$; the 40 lines then decompose as 1 (the Bell line itself) + 12 (lines meeting it — the gauge sector) + 27 (lines disjoint from it — the matter sector), with substrate reading $1 + k + q^q$. The vacuum decomposition that organizes the Standard-Model derivation is literally the photon's view of its own now-context: *self = the seed of self-entanglement.* The Bell-line stabilizer has order $51840/40 = 1296 = \mu^2 q^{q+1}$ — the same $1296 = 6^4$ that counts the compass D_{10}-incidence pairs of Theorem 12.4 (36 schedules $\times$ 36 core pairs), but Theorem 2.3 proves that this is not a regular-action identification. Instead, the projective Bell stabilizer resolves the incidence carrier as $648 + 324 + 162 + 162$, the exact mirror/schedule/left-cache/right-cache prefix decoder, while the point parabolic supplies two persistent 648 address sheets. The two cache words 110 and 111 are now resolved more finely by Theorem 10.6: they name two copies of the same H/C_4 cache, not two inequivalent internal chiral modules. Both double-cover one canonical H/D_8 base of 81 routes, and the complete cache quarter is an exact D_8-over-81 square bundle. This reversible address plane is provably not the non-split $81 \to 162 \to 81$ transport plane, despite their common 162 count. The companion also supplies the master equation at Hilbert-space level ($q^2 = q + q!$: the 9-dimensional past$\otimes$future space splits into 3 diagonal "now" histories and $6 = q!$ context-changing ones), and a ledger identity: temporal gate teleportation consumes exactly 2 classical trits of memory — the same 2-trit cost that prices the arrow of time in the self-loop fixed point of §11. Storing a state for one's future self and forgetting one's past cost the same two trits. The 27 itself now has a dual decoding: the 27 non-collinear *points* form a torsor under the genuine Heisenberg group (the normal 3_+^{1+2} of the point parabolic, acting simply transitively), whose invariant relations reconstruct the whole classical 27-zoo — the 9-triangle fibration, $\mathrm{GQ}(2,4) = \mathrm{SRG}(27,10,1,5)$, and the Schläfli graph $\mathrm{SRG}(27,16,10,8)$ of the E_6 27-lines world — while the 27 disjoint *lines* of the Bell shell form a torsor under elementary $\mathbb{F}_3^3$ (the line parabolic's O_3), a flat 3-trit register cube carrying *no* invariant strongly regular graph. Curvature for states, flatness for programs: the matter sector is a Heisenberg manifold on the state side and a ternary address space on the operational side. The register reading is exact: assigning each Bell-shell context its 3-trit address, the four pair relations $[4,4,6,12]$ are the difference-vector classes, and the geometry is a function of register arithmetic alone — meeting contexts ($\pm$-paired 4-classes) share exactly 1 transversal onto the Bell line, near-skew (12) exactly 2, far-skew (6) exactly 0: base-3 register incidence for matter contexts, twin to the charts' base-2 Gray arithmetic. *Witnesses:* `bt858_heisenberg_shell_torsors.py`, `bt860_bell_shell_register_arithmetic.py`.

Three rows deserve emphasis. The *Steinberg row* is the strongest protection statement available: the register is one irreducible representation, so *any* equivariant leakage operator is zero by Schur's lemma — protection by symmetry, not by redundancy. The *compass rows* say that what physics sees as delocalized icosahedral order, engineering sees as a navigation instrument: 12 needles per schedule, each carrying Petersen addressing, five-chart caches, and a dark-sector mirror. The *universal-polytope row* states the program's sharpest structural lesson: the hardware keeps the local physics of the exceptional objects and refuses their global rigidity; the refusal (an integer non-divisibility) is precisely why the architecture is distributed.

14.5 Beyond the laboratory

A different contract between theory and experiment. Every layer of this machine is specified by integers and verified by scripts that re-derive those integers from first principles in seconds. The cultural implication is a publishing model in which papers *run*: a claim is not a sentence but a reproducible computation, and a correction is a commit. This program has already operated under that contract (see the ethos below) — twice it corrected its own published claims at the source, and the corrected values were more natural than the originals. If the broader physics literature adopted executable claims at even a small scale, entire classes of irreproducibility would become type errors.

Quantum hardware without a building. Architectures that need millikelvin cryostats centralize by economics: only institutions can own them. A universal architecture whose parts list is catalog optics — source, splitters, delay fiber, modulator, detectors — decentralizes by the same economics. The realistic near-term role is not supremacy-class computation but *certified* small-scale quantum information: teaching laboratories that can verify 51840 and 36/40 on a bench; metrology nodes whose clock is protected by an irrational rotation; network testbeds where gates and routes are the same hardware. A machine whose benchmarks are exact integers is also the natural pedagogical machine: a student can check the whole specification.

Communication and computation stop being separate utilities. In this architecture there is no boundary at which computation hands off to networking — the gate groupoid and the transport groupoid coincide, and the fractal substitution makes the computer/network identity scale-invariant. The long-run implication, if any physical system realizes this pattern at depth, is infrastructural: bandwidth and compute become one provisioned quantity, scheduling becomes group theory, and the spectrum-allocation problem for the network's measurement contexts is solved by a finite library (36 timetables with a known switching metric) rather than by auction or contention.

Society-scale stakes of a zero-parameter physics. Separately from the machine, the substrate program's physics claims — 54+ observables, 14 falsifiable predictions, no adjustable parameters — carry a specific societal meaning: they are maximally exposed. A theory with no dials cannot retreat; the 2027 neutrino-hierarchy determination, the proton-decay window, the CMB measurements will each either hit the predicted integers or end the corresponding pillar. Public trust in fundamental science is built by exactly this posture — predictions that can die — and the machine half of the program extends the posture to hardware: the device specification is its own audit.

What could go wrong, said plainly. The architecture is exact; its physical realization is not yet demonstrated beyond the component level. The fractal scaling laws are combinatorial theorems, not engineering schedules. The security readings are directions, not proofs. And the identification of the substrate with fundamental physics stands or falls with the prediction table, not with the elegance of the mathematics. This paragraph exists because the program's ethos (below) requires the failure modes to be listed next to the claims.

14.6 The ethos

Several pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3,3)$ is 7, not 10 (no ovoid exists; the "perfect graph" block is withdrawn); the Kochen–Specker optimum is 36/40, not 34/40; the pentad-sharing count of BT844 was corrected by the chiral-orbit census of BT845 within one working session; and the tomotope's framing was corrected from "non-polytopal maniplex" to "uniform polytope with non-polytopal minimal covers" when the literature was read precisely; and the "$\mathbb{Z}_3$ Berry phase as a discrete second Chern class" was corrected (BT871) — the uniform Berry 2-cochain is a coboundary over $\mathbb{F}_3$, so the physical $\mathbb{Z}_3$ is the order-3 action on the Steinberg register, not a curvature class. In every case the corrected value is *more* substrate-natural than the claim it replaced ($\alpha = \Phi_6$; budget $= (q!)^2/v$; $432 = 216 + 216$ chiral; infinitely many covers as the actual pathology; the $\mathbb{Z}_3$ as a group action rather than a cocycle) — the pattern a real structure shows when probed honestly. The program treats corrections as first-class results: they are committed, logged, and cited like theorems, because a corpus that cannot heal is a corpus that cannot be trusted.

A Verification ledger

Every theorem above is backed by an executable script in `analysis/` with results in `data/`; the chain BT739–BT839 was developed and pushed incrementally with exact arithmetic (GF(p) elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

layer	primary witnesses
duality / carriers	bt817, bt820, bt821
atlas / building / web	bt773, bt777, bt744, bt779
mirror middleware / tomotope runtime	bt814, bt815, bt826
braids / registers	bt740, bt741
vacua / geography / transitions	bt808–bt813, bt816
fuel / contextuality	bt818, bt822, bt823
memory / Steinberg	bt742, bt744
clocks / quasicrystal / arrow	bt774, bt819, bt820, bt823
sync / schedules / overlap law	bt817, bt819, bt835
Grünbaum–Coxeter hemicells / library geometry	bt836, bt837, bt839
two compasses / $4 \cdot K_{10} = 12$ Petersens / chart double cover	bt843–bt845
pentad anatomy / schedule reconstruction / dark dodecahedra	bt846, bt847
amalgamation ladder / universality / J_1	bt848
maniplex barren rung / class $2_{\{0,1,2\}}$ / Cayley over V_4	bt849–bt851
seventeen universals / Aut(tomotope) universal	bt852
dark orbit zoo / shadow bijection / K_5-complete dark sector	bt853–bt855
dark charts = mirror bus (G-set iso)	bt856
Heisenberg/$\mathbb{F}_3^3$ shell torsors, Schläfli decoding	bt858
Bell-shell register arithmetic ($[4,4,6,12]$ named)	bt860
code register = Steinberg ($[[240,81,4,3]]_3$)	bt861
H_2 = sign-twisted lines (homological dictionary)	bt862
three generations = Steinberg vanishing	bt863
triality census (matter-blind, gauge-visible)	bt864
joint generation $\times$ chirality grading	bt868
chirality = polar-pair involution ($45 + 36$)	bt869
spanning-tree gravity $\tau = 2^{81}5^{23}$	bt870
Berry-phase correction ($\mathbb{Z}_3$ is the Steinberg action)	bt871
Ihara zeta = spanning-tree gravity bridge	bt872
two-zeta duality / walk census	bt873
texture triality = long-root transvection	bt874
Yukawa rule = $\mathbb{Z}_3$ grade conservation	bt875
gauge group $1 \oplus 3 \oplus 8$ from $C(R)$	bt876
gauge parity = duality ($A_4 \to S_4$)	bt877
generation charge-conjugation ($N/C = \mathbb{Z}_2$)	bt878
flavor group S_3 ($\mathbb{C}[27] = 6{\cdot}1{+}3{\cdot}1'{+}9{\cdot}2$)	bt879
generations = centre of the gauge group	bt880
spacetime = space of 40 local gauge groups	bt881
gauge connection flat on edges, $2T$-curved on Q	bt882
curvature 2-form (quaternionic, on Q)	bt883
gauge flux (Wilson loops, $\mathbb{Z}_3$/lines, $\leq 12/Q$)	bt884
dual-torsor Steinberg compiler / native generation flag	bt865
oriented H_2 irreducible spectrum / outer lift	bt866
cache D_8-over-81 fibration / split–non-split boundary	bt867
chirality bookkeeping / Bell–compass parabolic router	bt857, bt859
universality	bt825
fractal computer/network	bt827
runtime compiler / sentinel / commit / cover boundary	bt828–bt834, bt838
corpus corrections	bt818, bt823, bt824

References

[1] W. Dahn, *The $W(3,3)$ Substrate: Master Manuscript* (`w33_paper.tex`), Theory of Everything Project (2026) — master equation $q! = 2q$, parameter closure, Bose–Mesner field equations, and the full observable derivation chain.

[2] W. Dahn, *Self-Entanglement Companion*, Theory of Everything Project preprint (2026).

[3] M. Howard, J. Wallman, V. Veitch, J. Emerson, Contextuality supplies the 'magic' for quantum computation, *Nature* **510**, 351 (2014).

[4] P. T. Dumitrescu *et al.*, Dynamical topological phase realized in a trapped-ion quantum simulator, *Nature* **607**, 463 (2022).

[5] D. Gottesman, I. Chuang, Demonstrating the viability of universal quantum computation using teleportation, *Nature* **402**, 390 (1999).

[6] Y. Saad and M. H. Schultz, Topological properties of hypercubes, *IEEE Transactions on Computers* **37**, 867 (1988).

[7] A. Smith, Universality of Wolfram's 2, 3 Turing machine, *Complex Systems* **29**, 1 (2020).

[8] S. Wolfram, The Wolfram 2,3 Turing Machine Research Prize, https://www.wolframscience.com/prizes/tm23/.

[9] I. Niven, *Irrational Numbers*, MAA (1956).

[10] J. A. Thas, Ovoids and spreads of finite classical polar spaces, *Geom. Dedicata* **10**, 135 (1981).

[11] B. Monson, D. Pellicer, G. I. Williams, The tomotope, *Ars Mathematica Contemporanea* **5**, 355–370 (2012).

[12] T. Breuer, *GAP Character Table Library: $U_4(2) \cong$ PSp$(4,3)$*, ATLAS-derived maximal-subgroup data, online ATLAS table.

[13] B. McMillan, Two inequalities implied by unique decipherability, *IRE Transactions on Information Theory* **2**(4), 115–116 (1956), doi:10.1109/TIT.1956.1056818.